



Department of Anatomy, Histology and Embryology

Semmelweis University, Faculty of Medicine

Development of the veins

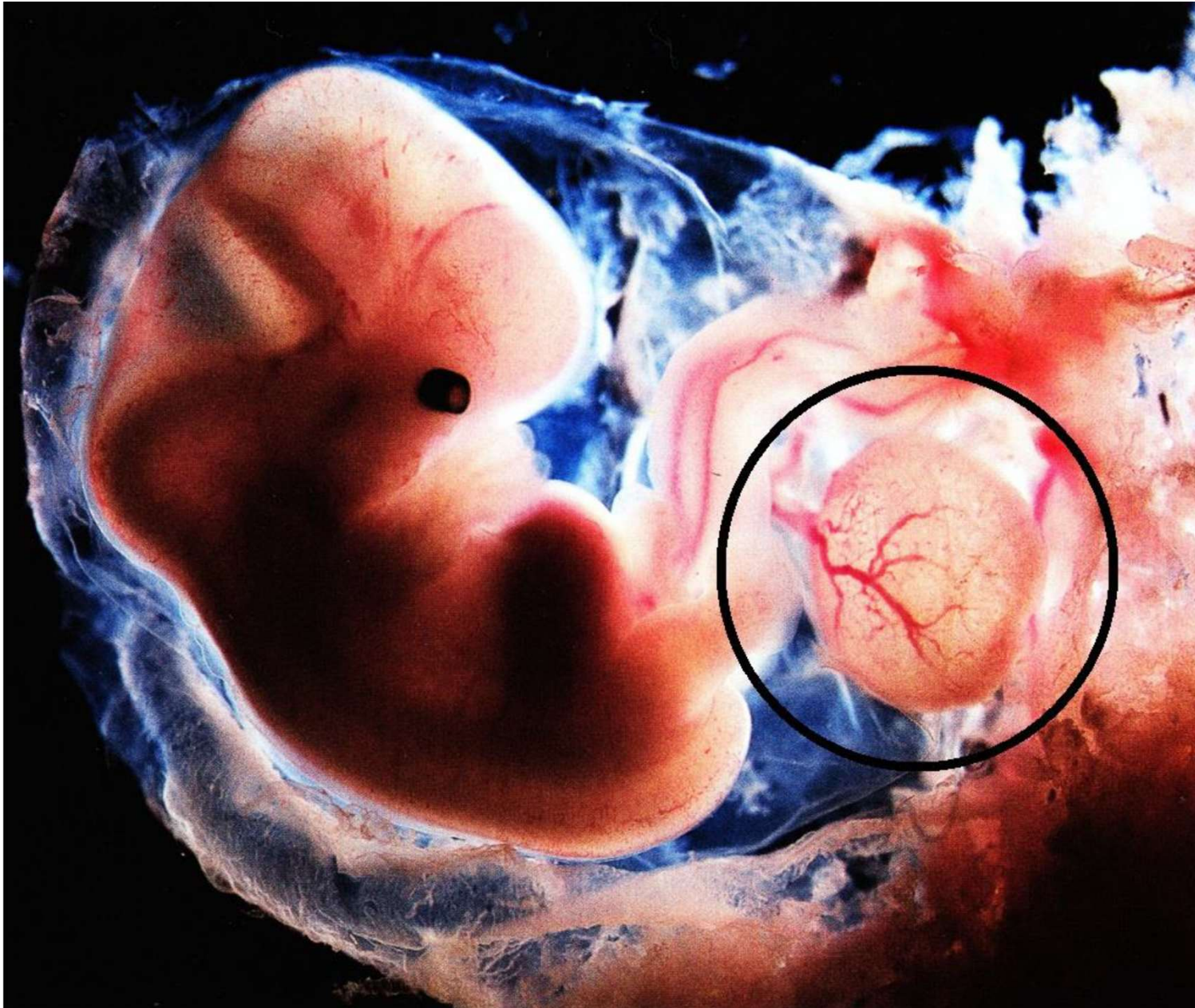
Dr. Németh Péter László

MD DMD DDS

2026

ANGIOGENESIS

Occurs during development



ANGIOGENESIS Occurs in adult tissue as well

- During wound healing



**Angiogenesis in
uterine lining**



**Angiogenesis in tissue
during wound healing**

VEGF (Vascular Endothelial Growth Factor)

Discovered in 1989. Role: endothelial cell proliferation, differentiation, migration and proper vessel patterning.

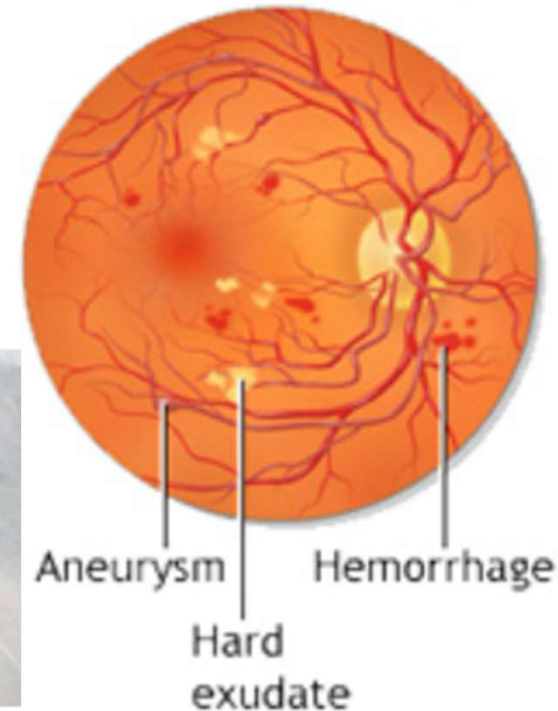
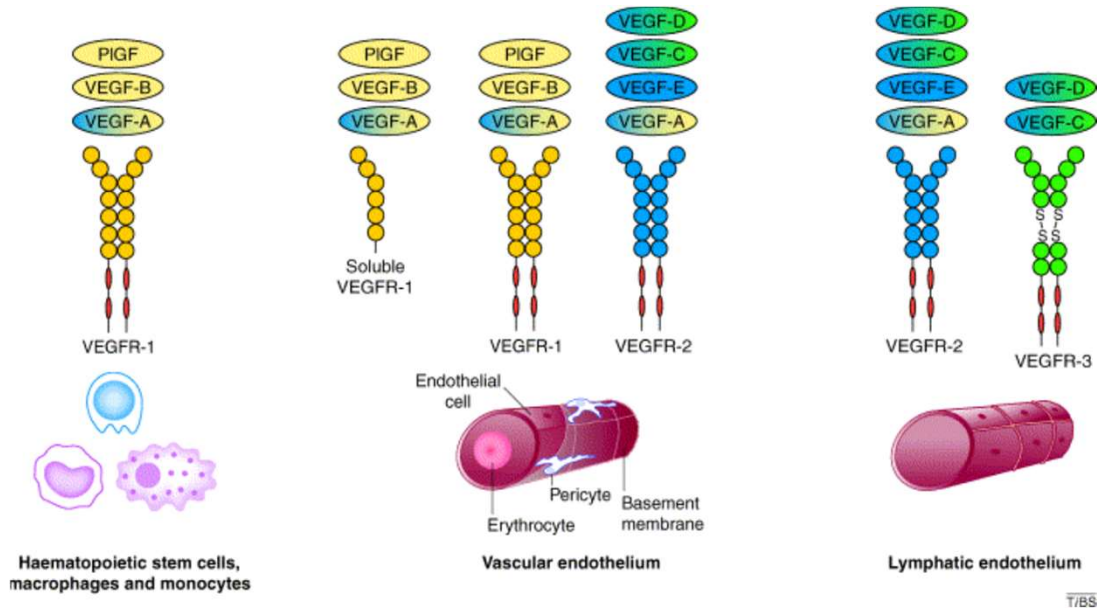
A heparin binding growth factor released by many tissues.

Basis of tumor/tissue neovascularization.

Binds to **tyrosine kinase active receptor**.

HIF-1a inducible – response to hypoxia

In the working-age population, it is the most common cause of blindness.



ranibizumab, aflibercept, bevacizumab, faricimab

<https://www.reviewofoptometry.com/news/article/sixyear-data-on-antivegf-impact-from-the-iris-registry>

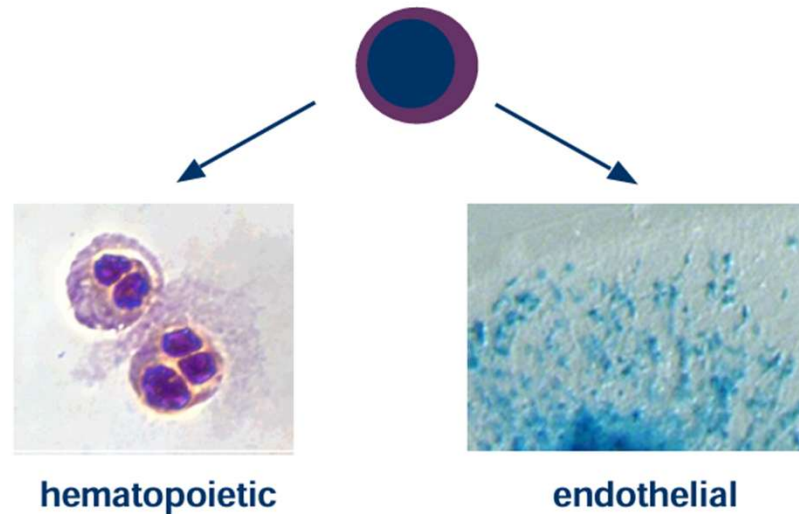
+ growth of abnormal vessels

<https://www.corkeyeclinic.ie/diabetic-retinopathy>

„Data from avian, mouse, and human embryos consistently show a close association between the endothelial and hematopoietic compartments. These findings suggest that a common precursor or hemangioblast is conserved between species during early development.”

Bailey, A. S., and W. H. Fleming, 2003: Converging roads: evidence for an adult hemangioblast. *Exp. Hematol.* 31, 987–993.

The hemangioblast



Hypothetical common progenitor for hematopoietic and vascular endothelial cells

Vasculogenesis

Coalescence of
haemangioblasts
[Cardinal veins]

1. Splanchnic mesoderm >
Haemangioblasts > **Blood islands**
[angioblasts + inner BPCs]
2. Angioblasts > **Endothelial cells**
3. **PRIMARY CAPPILARY PLEXUS**

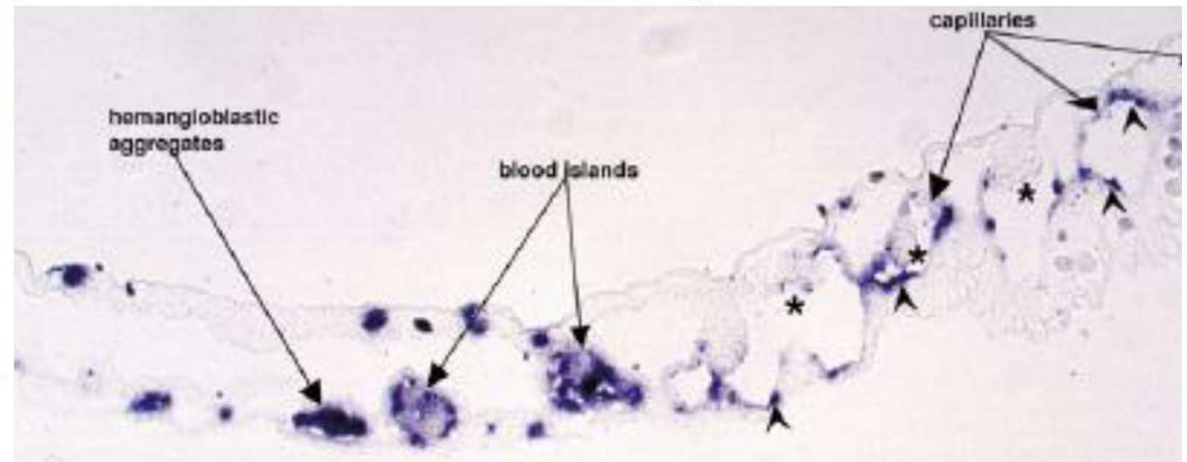
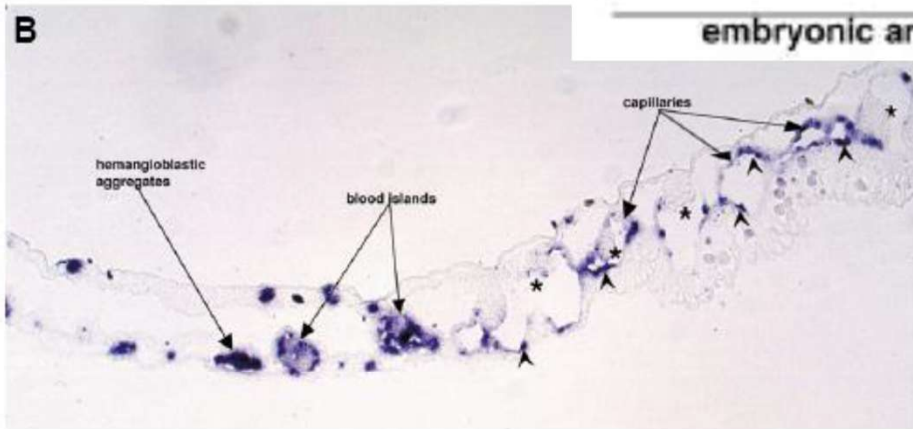
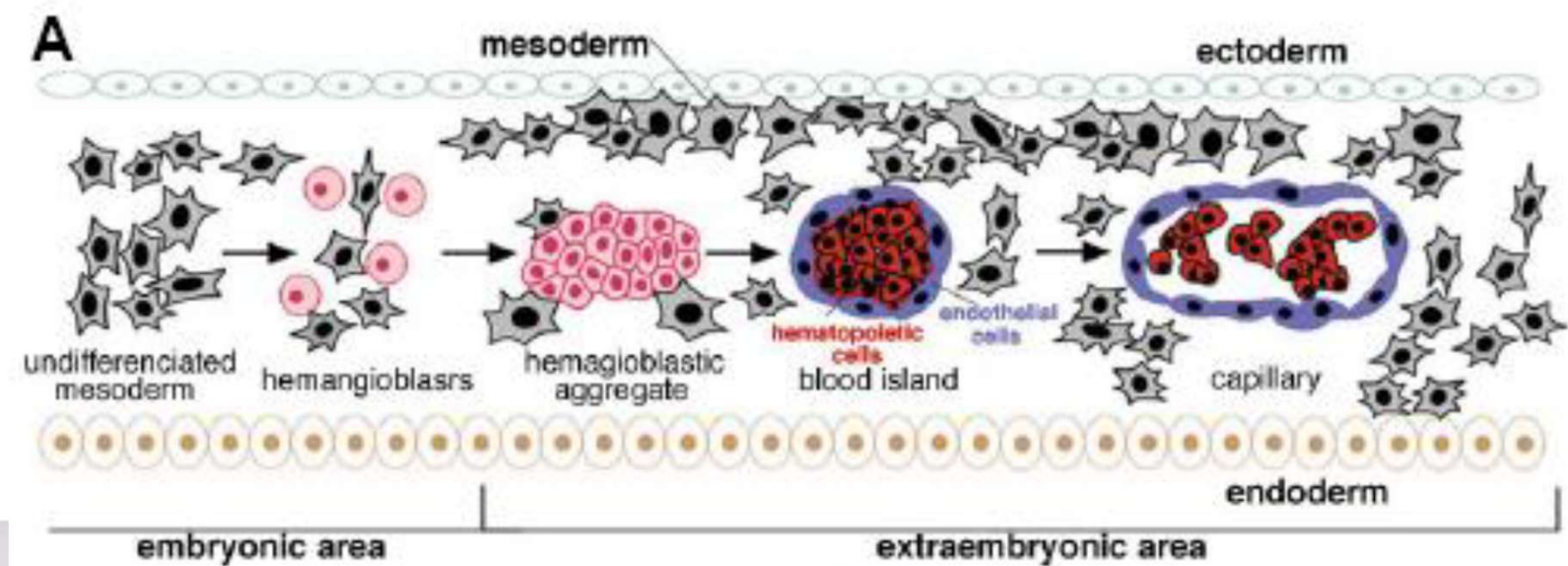
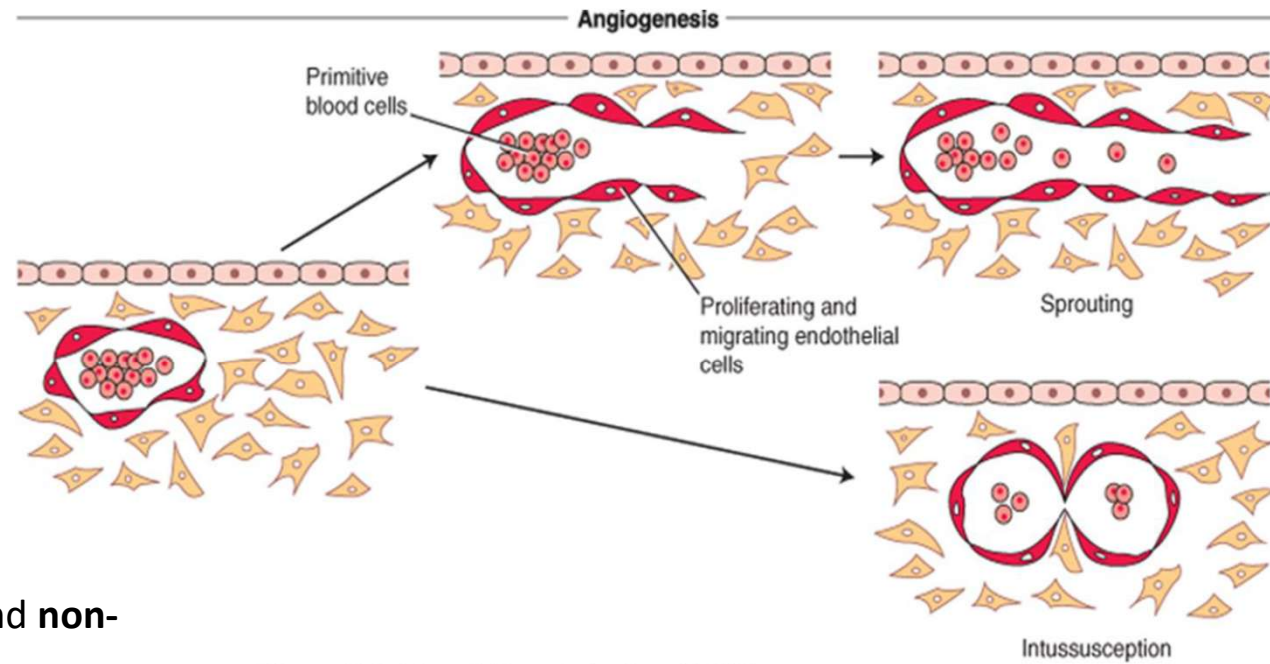
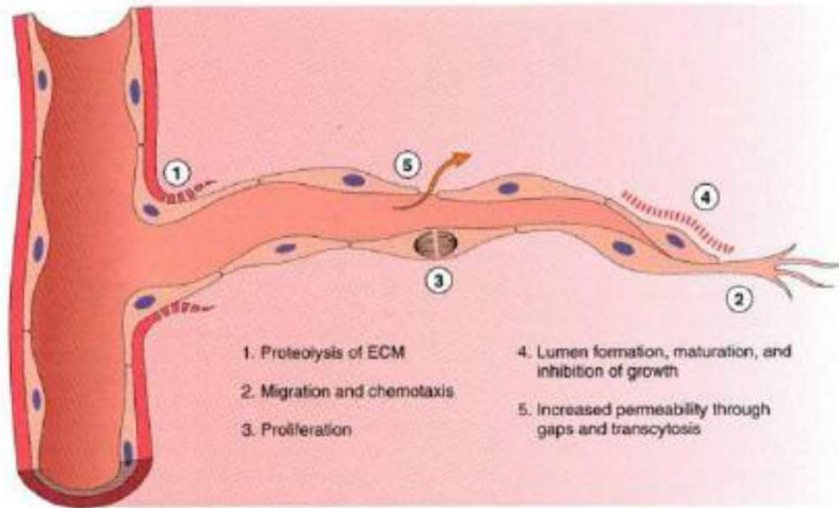


Fig. 3. Differentiation of yolk sac blood islands. (A) Schematic representation: see text for details. (B) Transverse section through the extraembryonic area of a 15-somite stage embryo, labeled with an antisense riboprobe specific for VEGFR2. Hemangioblastic aggregates are positive. In the blood islands and capillaries, expression of the receptor decreases in the hematopoietic cells (asterisks), but is maintained in the endothelial cells (arrowheads). Bar, 75 μ m.

Angiogenesis – Sprouting from existing vessels [Other veins]

Angiogenesis



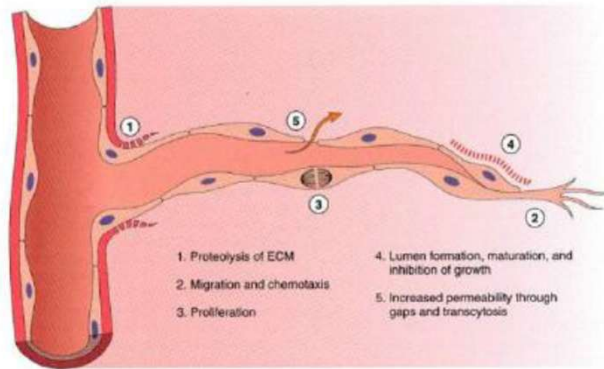
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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In reference to angiogenesis, **Sprouting angiogenesis** and **non-sprouting angiogenesis = intussusception** can be distinguished.

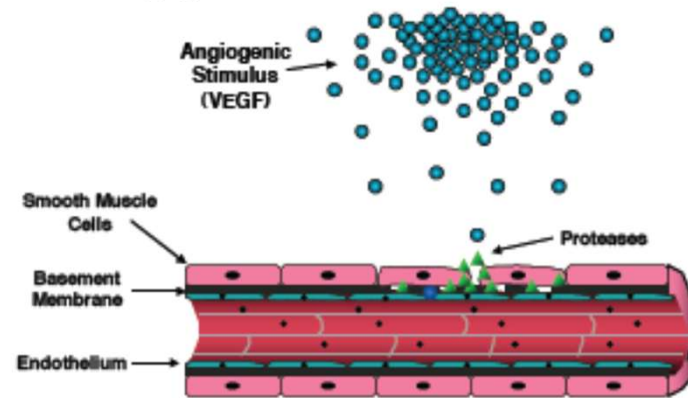
Kalka, C., T. Asahara, W. Krone, and J. M. Isner, 2000: [Angiogenesis and vasculogenesis Therapeutic strategies for stimulation of post-natal neovascularization]. Herz 25, 611–622.

Angiogenesis – Sprouting from existing vessels [Other veins]

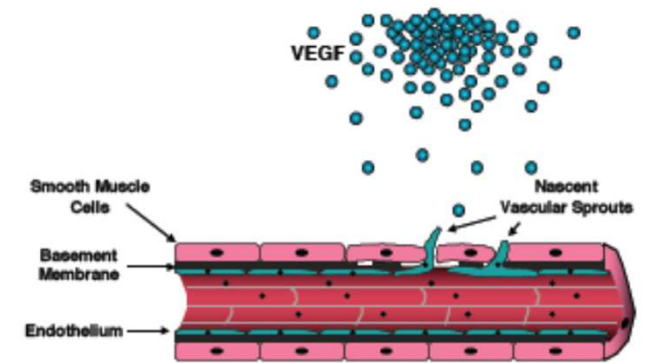
Angiogenesis



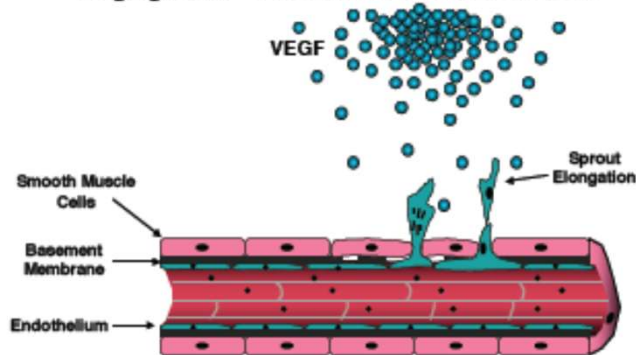
Angiogenesis - Basement Membrane Breakdown



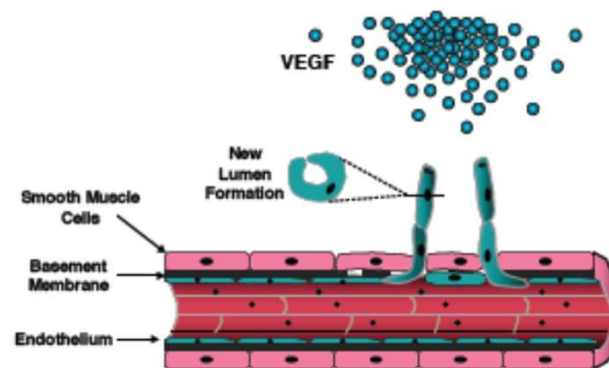
Angiogenesis - Endothelial Cell Migration



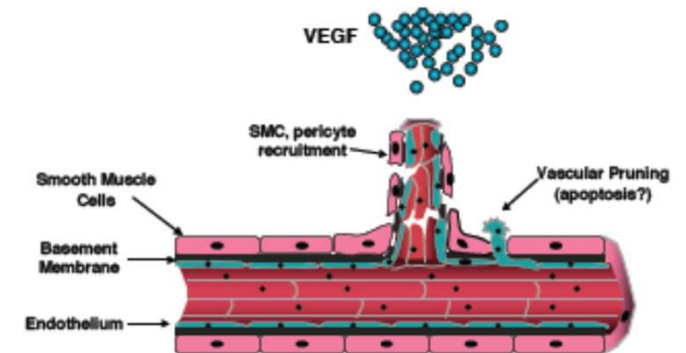
Angiogenesis - Endothelial Cell Proliferation



Angiogenesis - Capillary Morphogenesis



Angiogenesis - Vascular Maturation

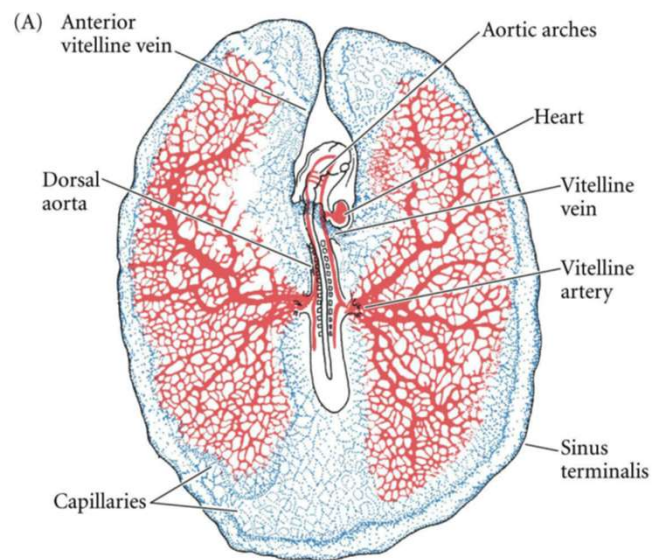


On the 4th week

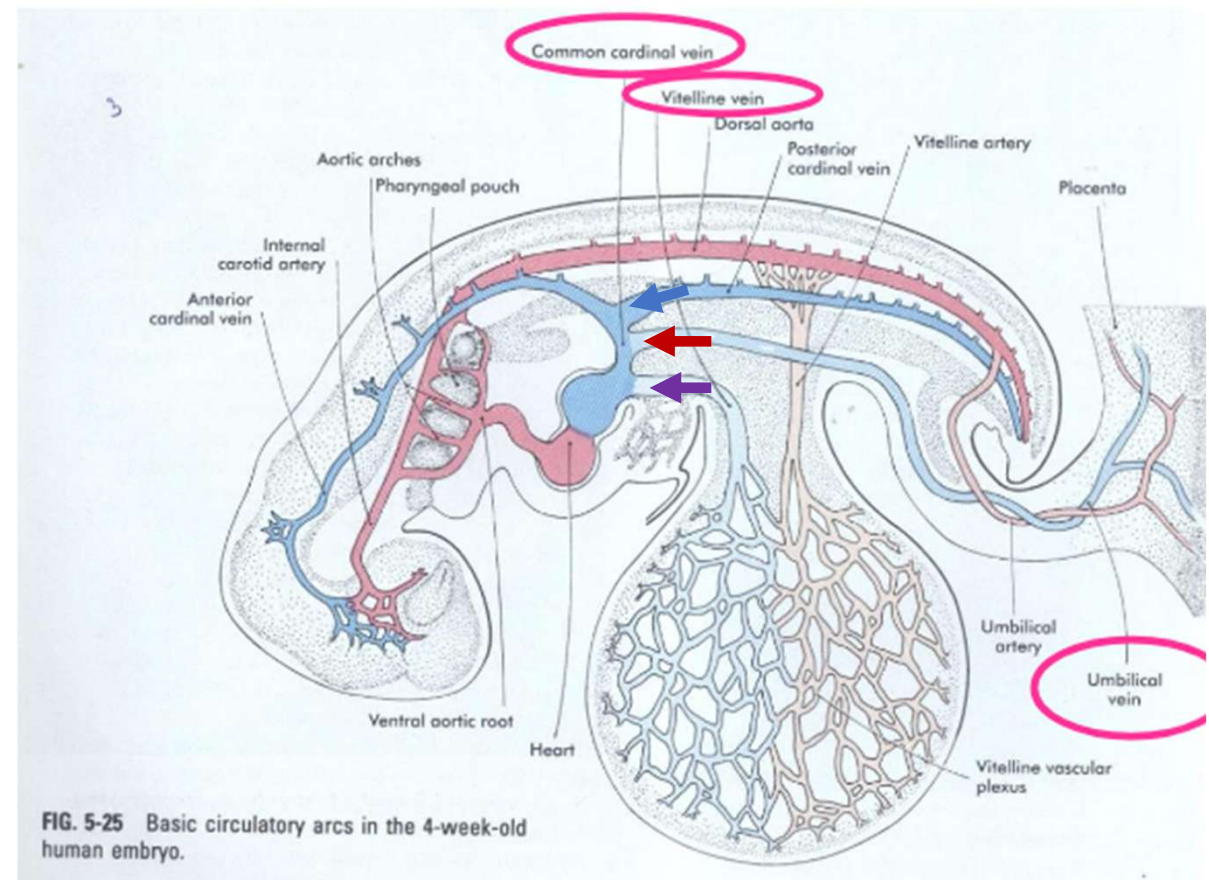
2 **Common cardinal veins** (Ducts of Cuvier) < **BODY O₂-**

2 **Umbilical veins** < **PLACENTA O₂+**

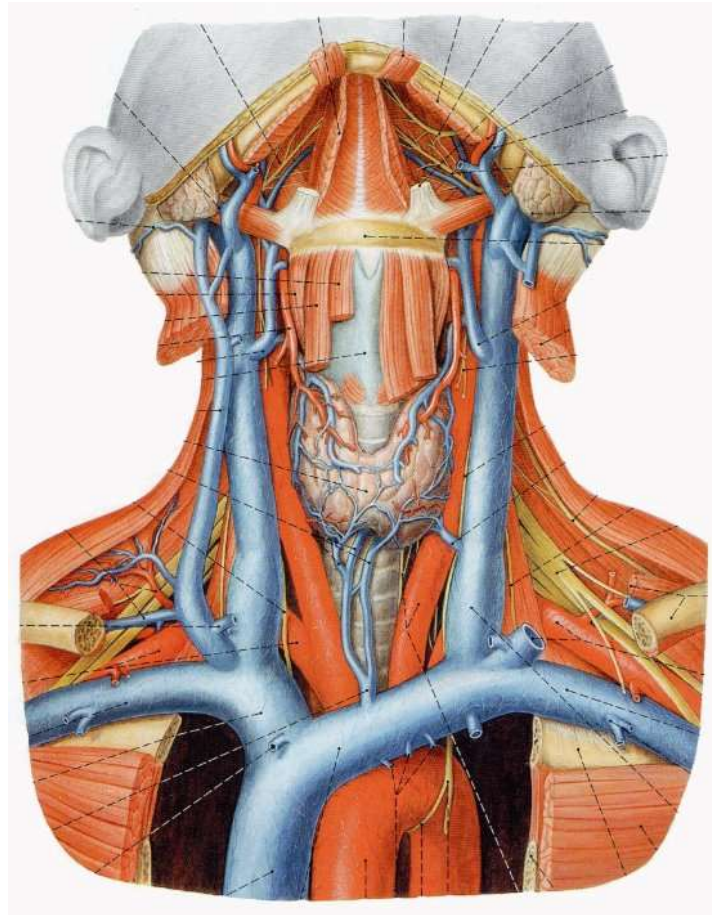
2 **Vitelline** (ophalomesenteric) **veins** < **YOLK SAC ~O₂**

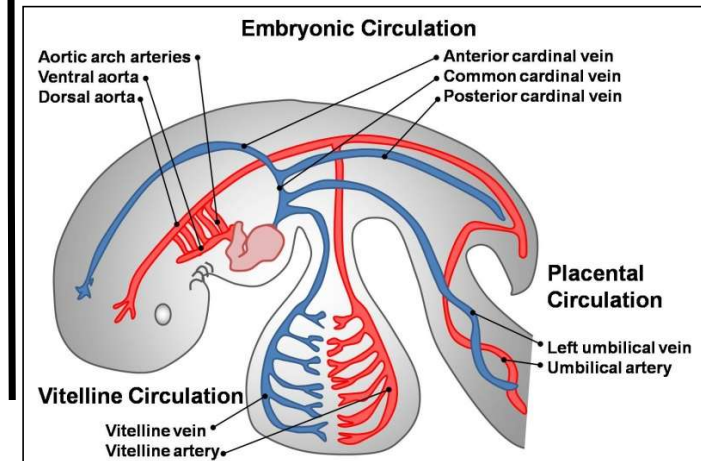
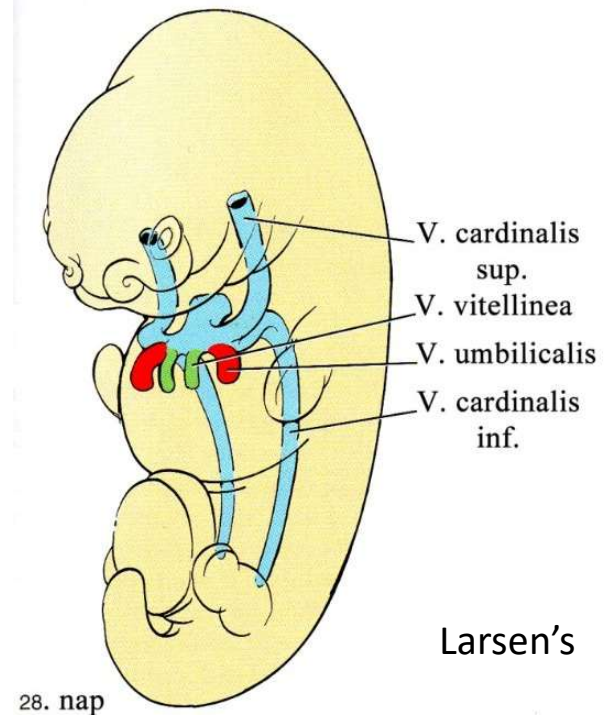
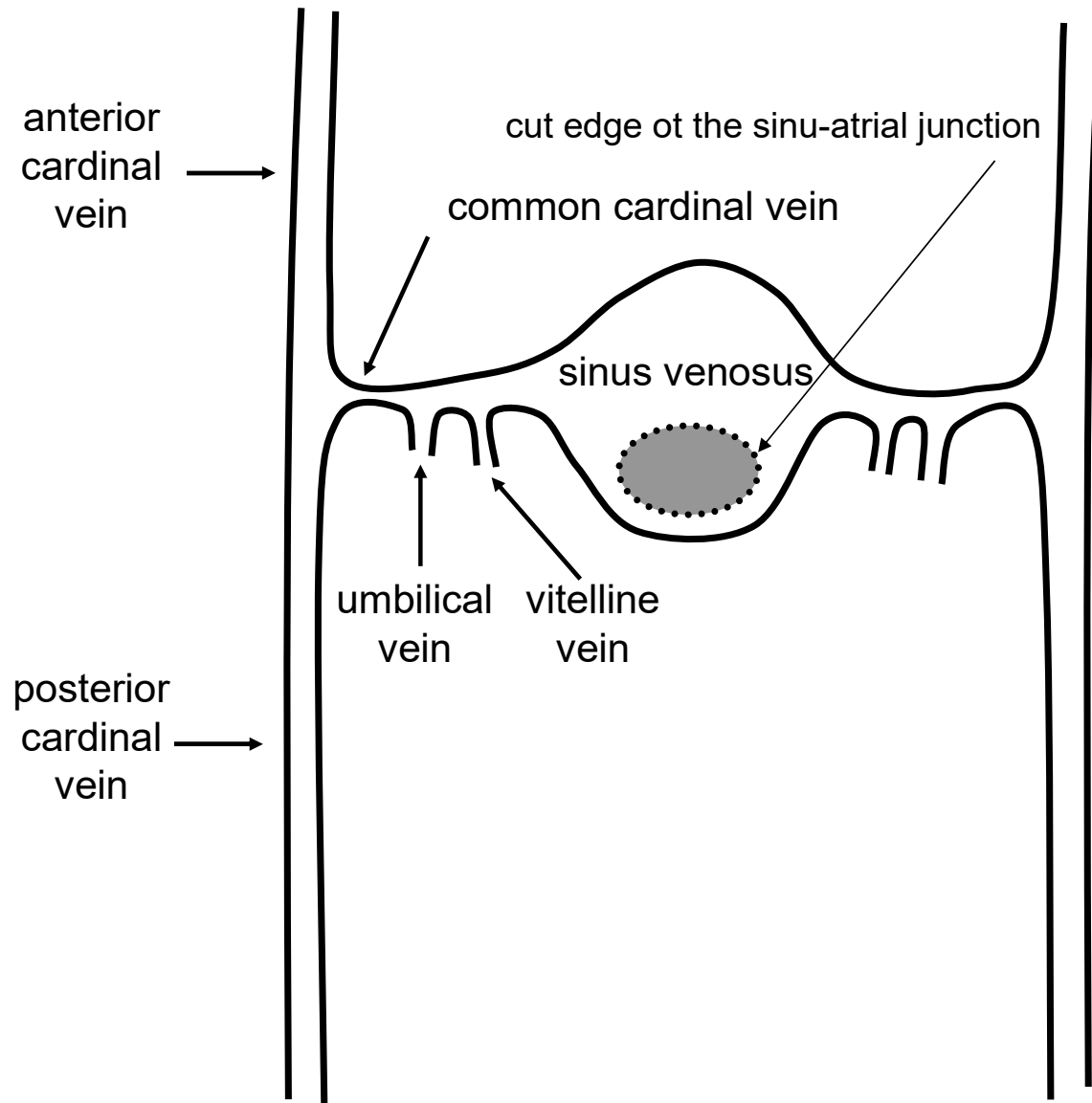


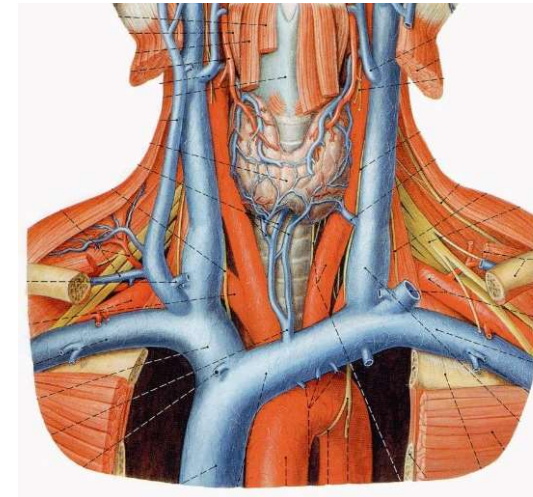
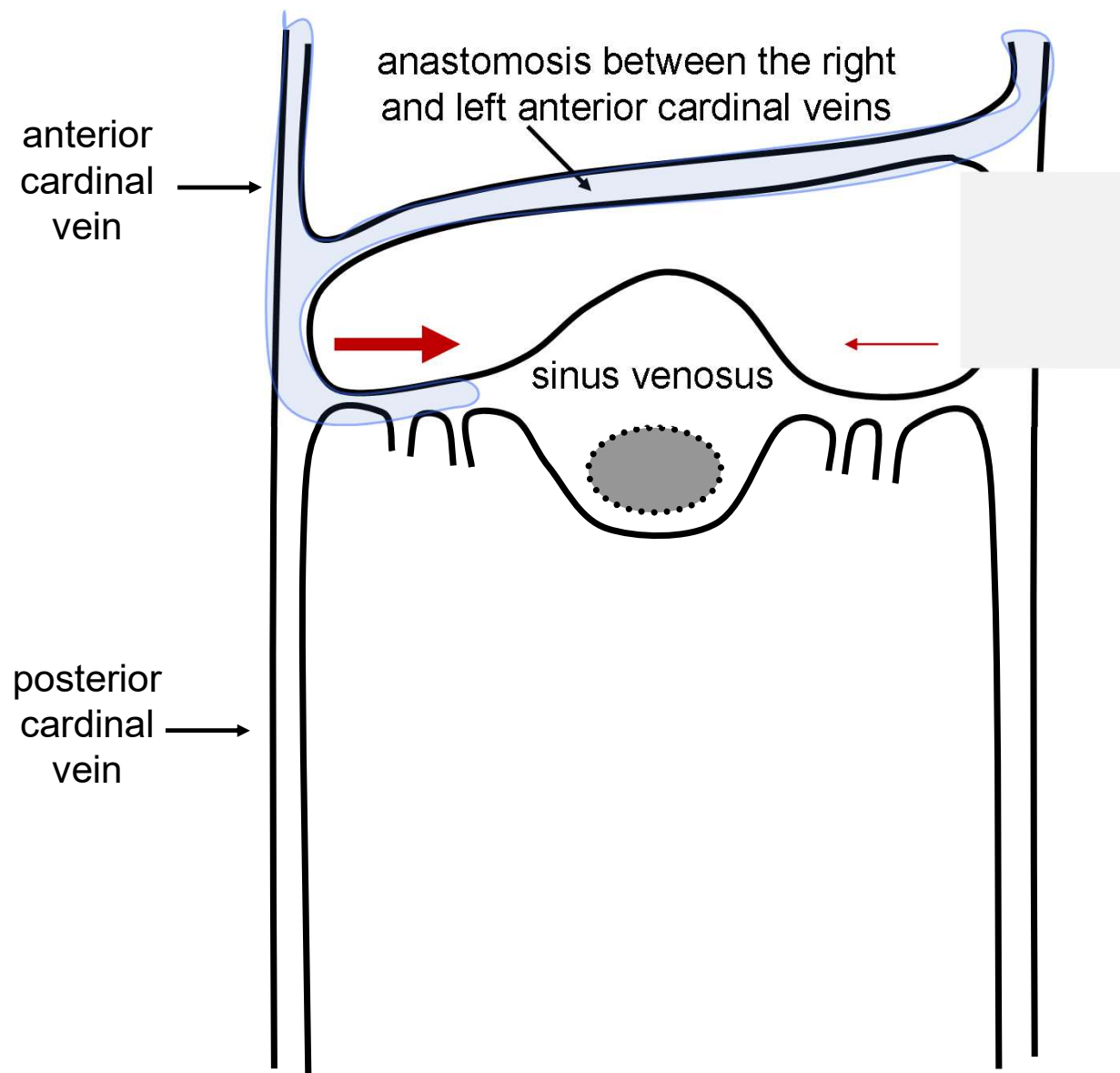
DEVELOPMENTAL BIOLOGY, Eighth Edition, Figure 15.9 (Part 1) © 2005 Sinauer Associates, Inc.



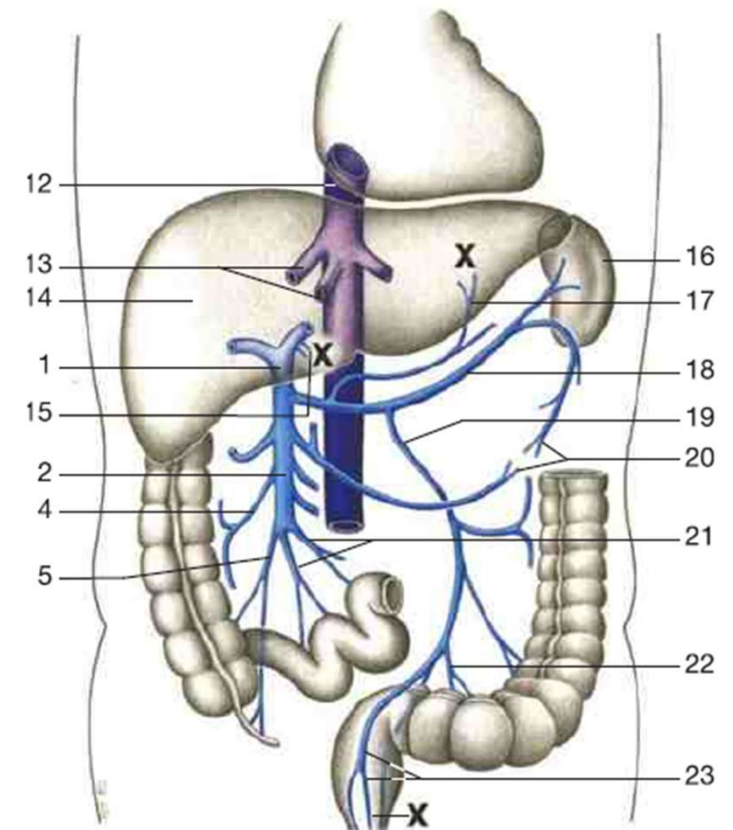
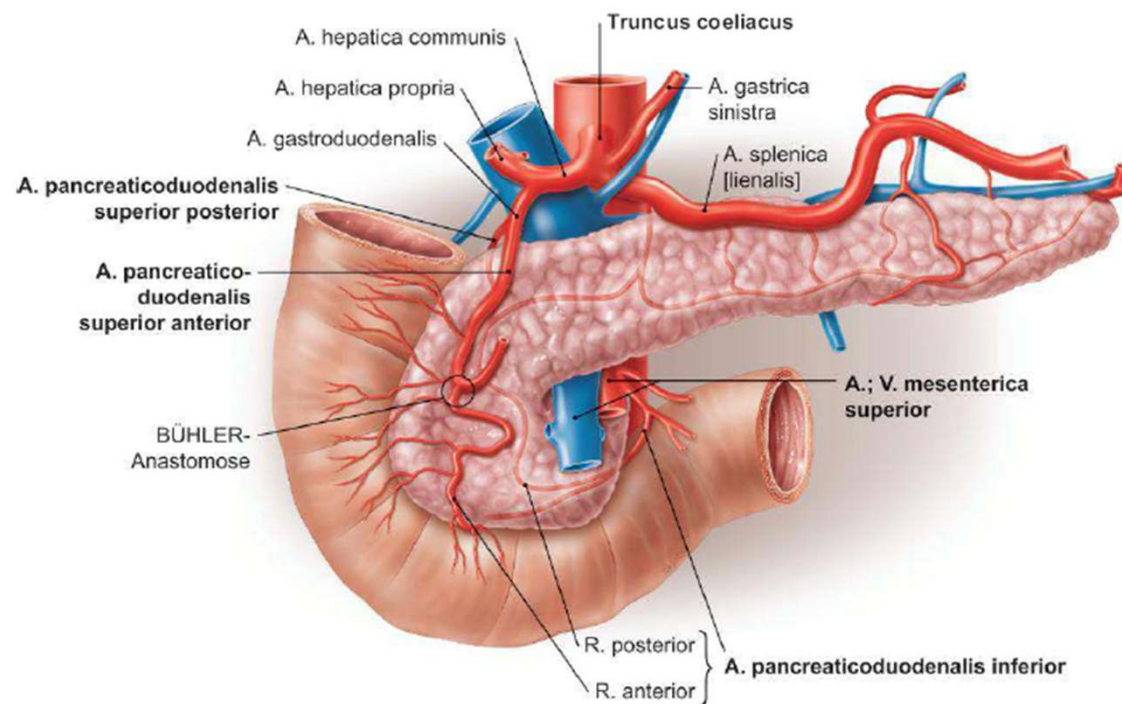
SUPERIOR VENA CAVA

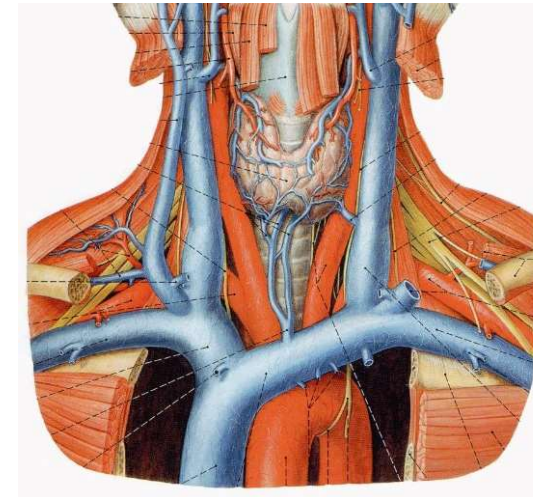
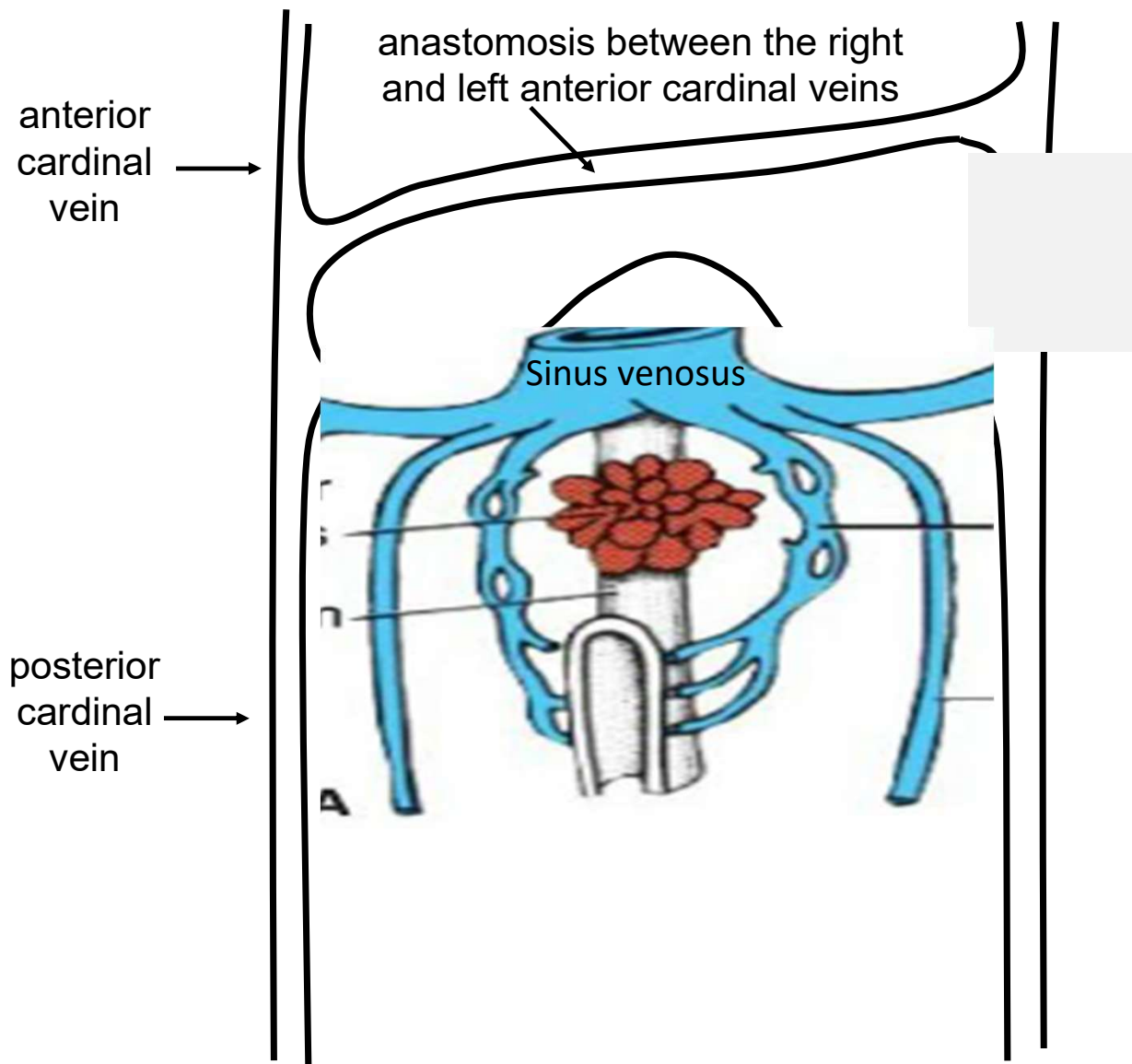




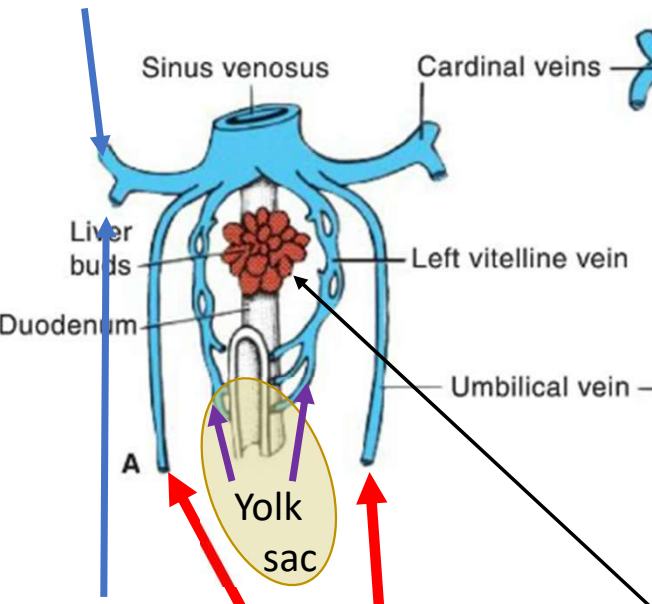


PORTAL SYSTEM



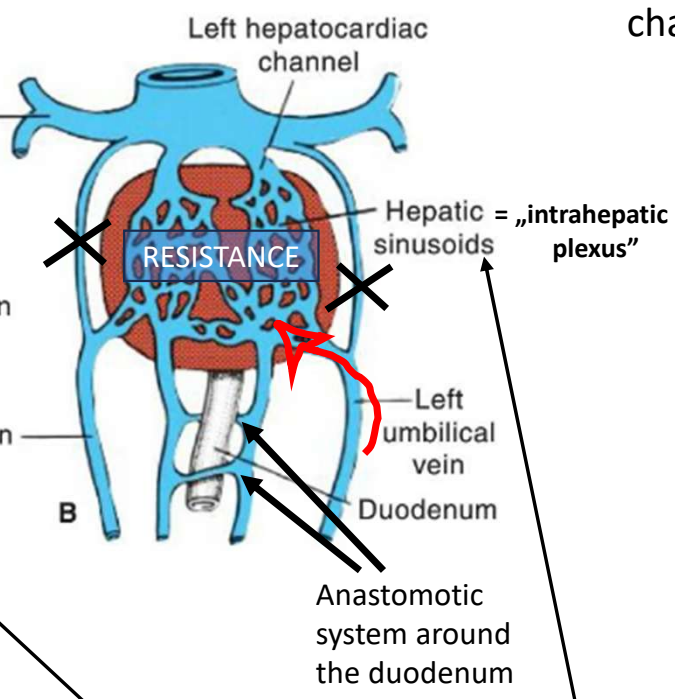


Upper body



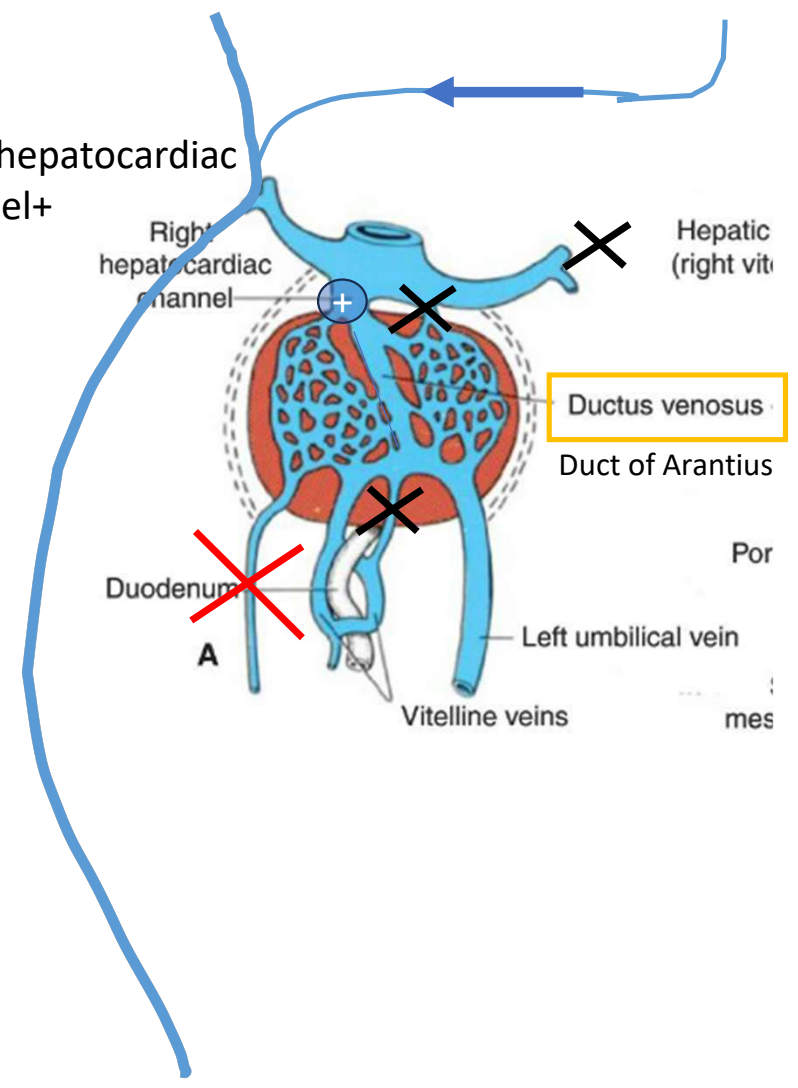
Lower body

Placenta

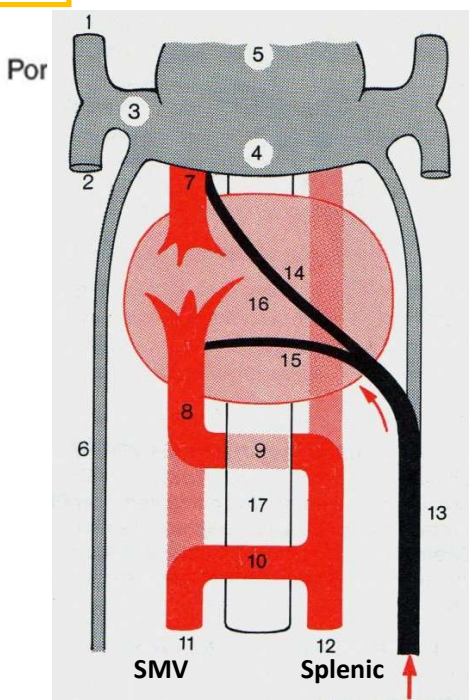
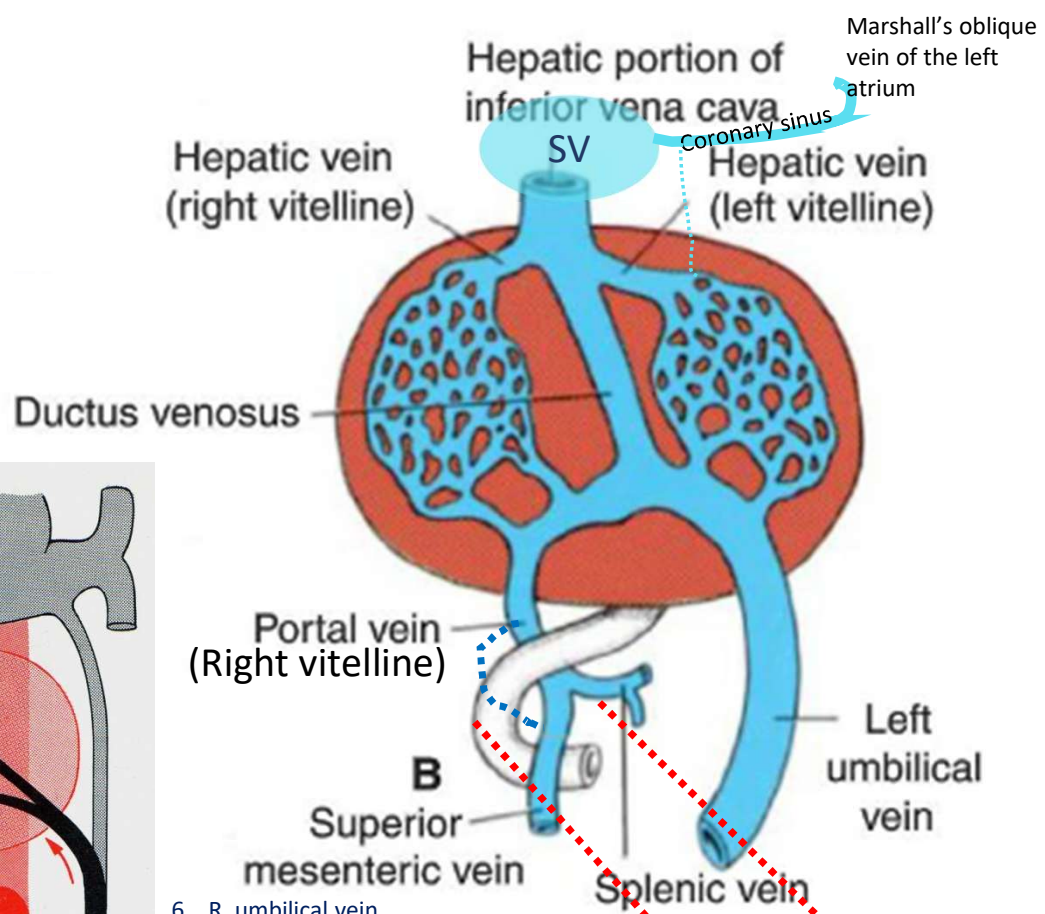
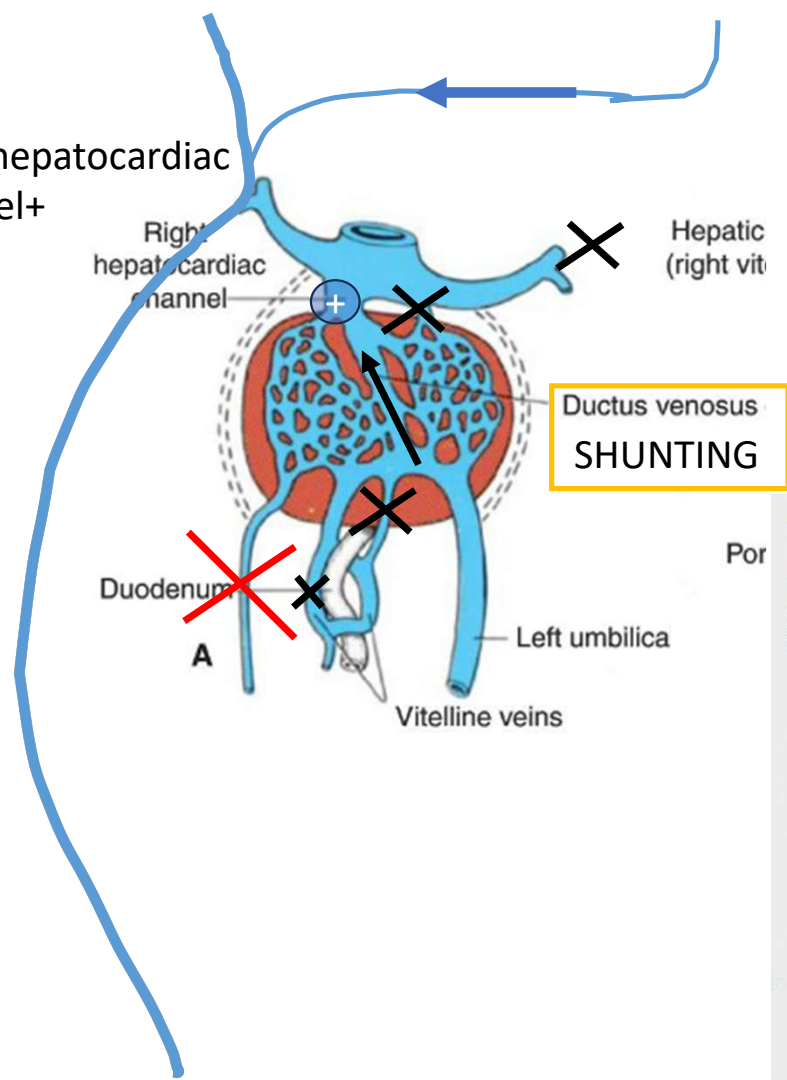


RAPIDLY PROLIFERATING LIVER
Into the septum transversum

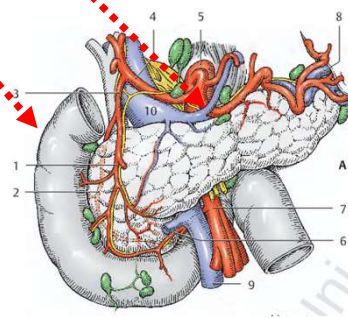
Right hepatocardiac channel+

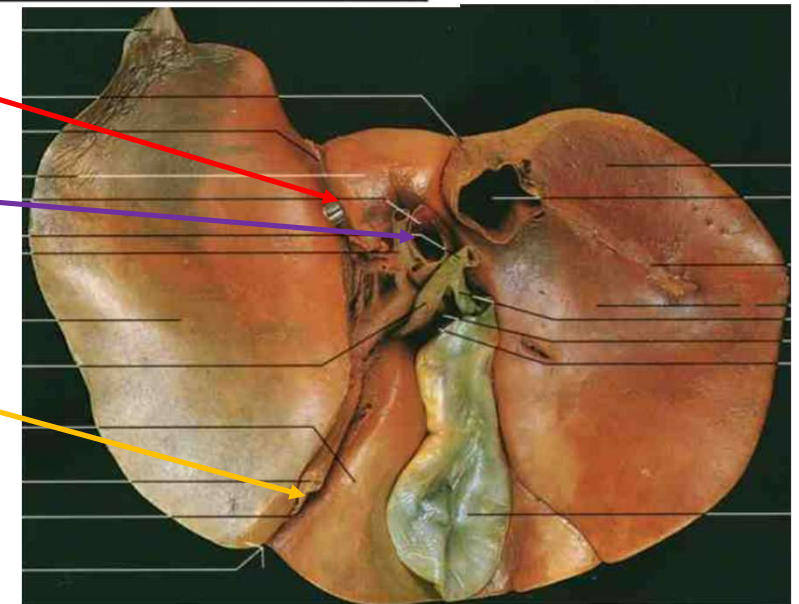
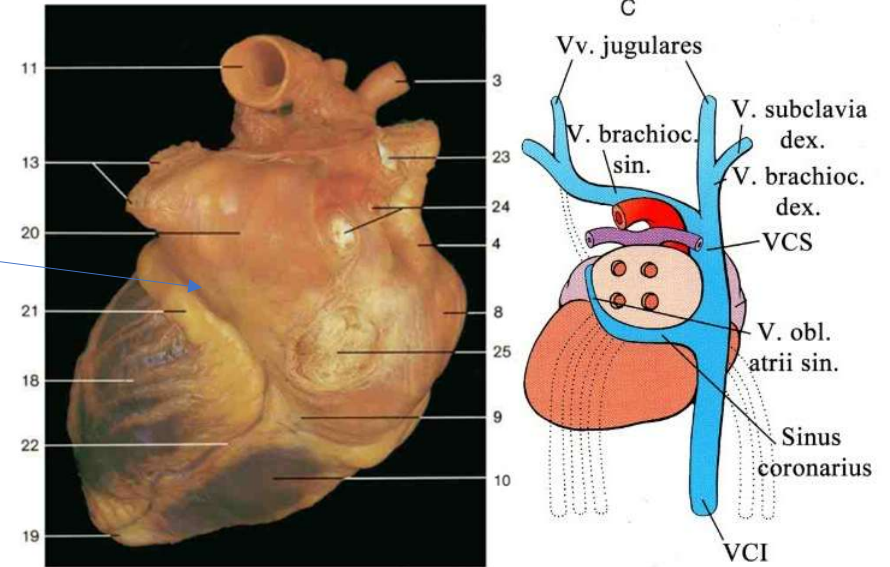
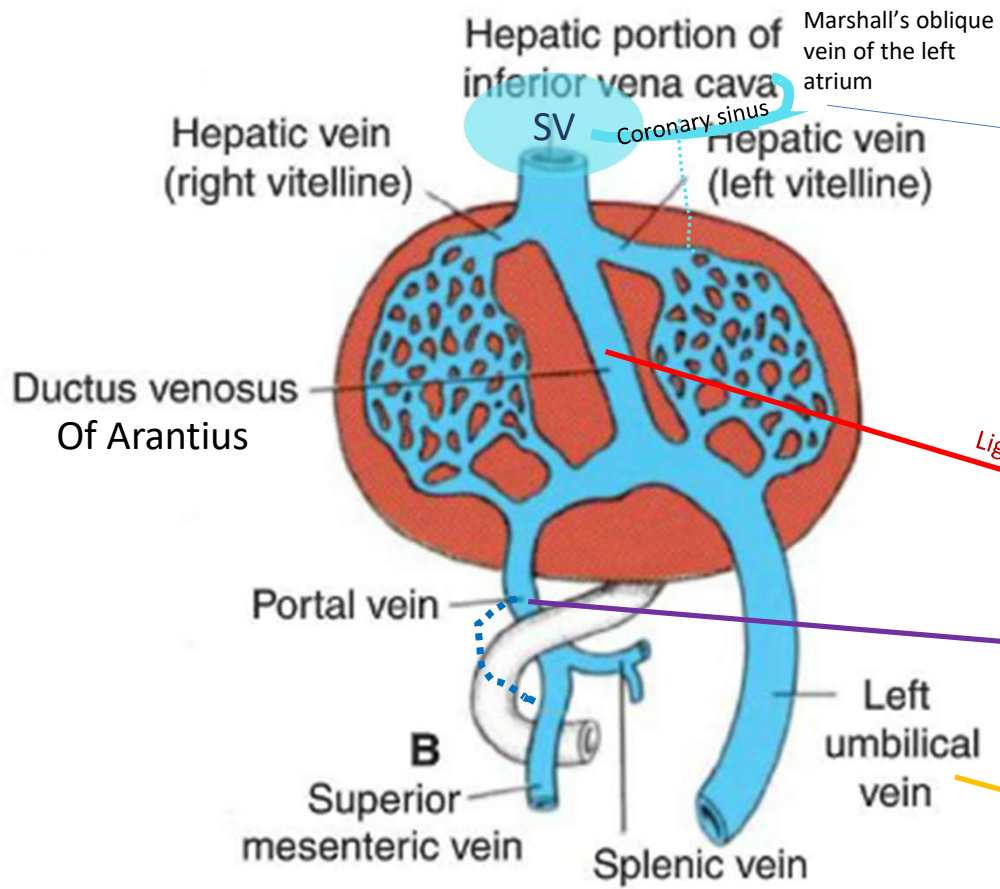


Right hepatocardiac channel+

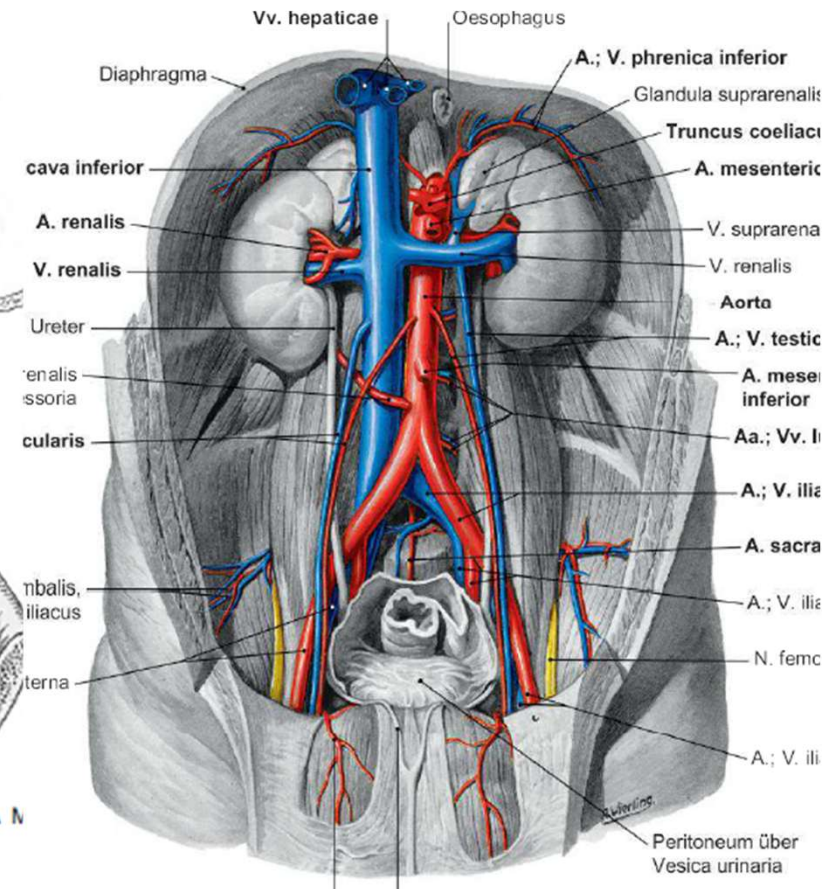
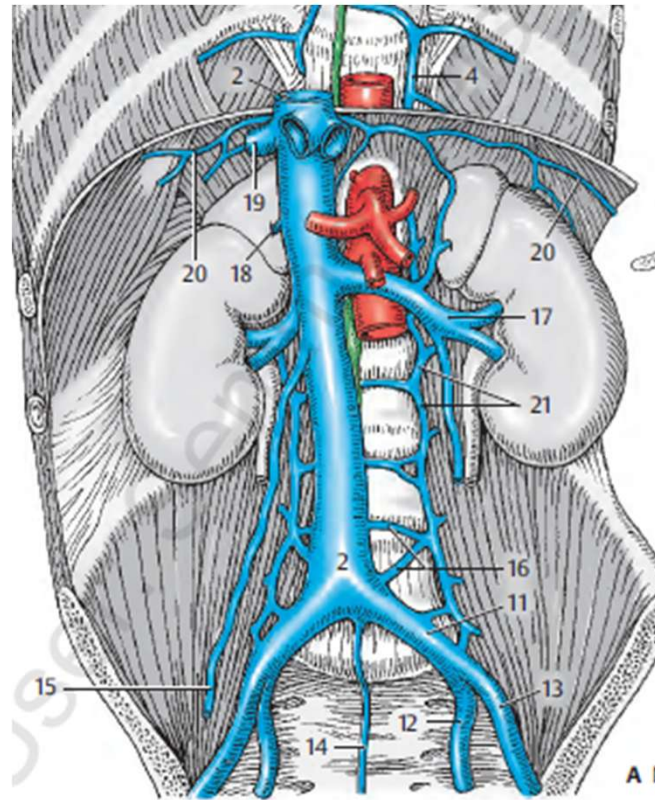
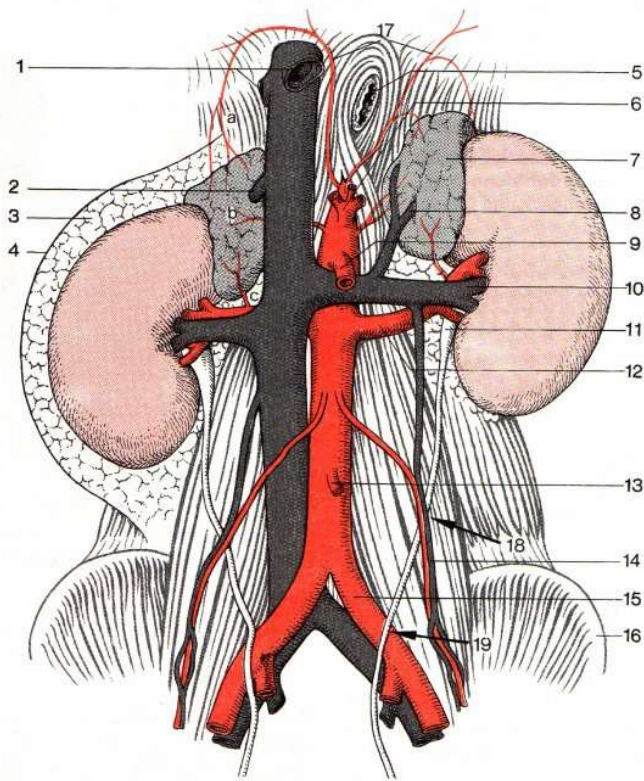


- 6. R. umbilical vein
- 13. Left umbilical vein
- 12. Left vitelline vein. >> **Splenic vein**
- 11. Right vitelline vein. >> **Superior mesenteric vein**
- 8. continuation (advehens hepatis vein = **PORTAL**)
- 7. cont. (revehens hepatis vein = **Hepatic veins**)
- 17. Duodenum
- 10. preduodenal anastomosis
- 9. retroduodenal anastomosis
- 15. Later > **Round ligament of the liver**
- 14. **Duct of Arantius (Lig.venosum) G.C.Aranzi**

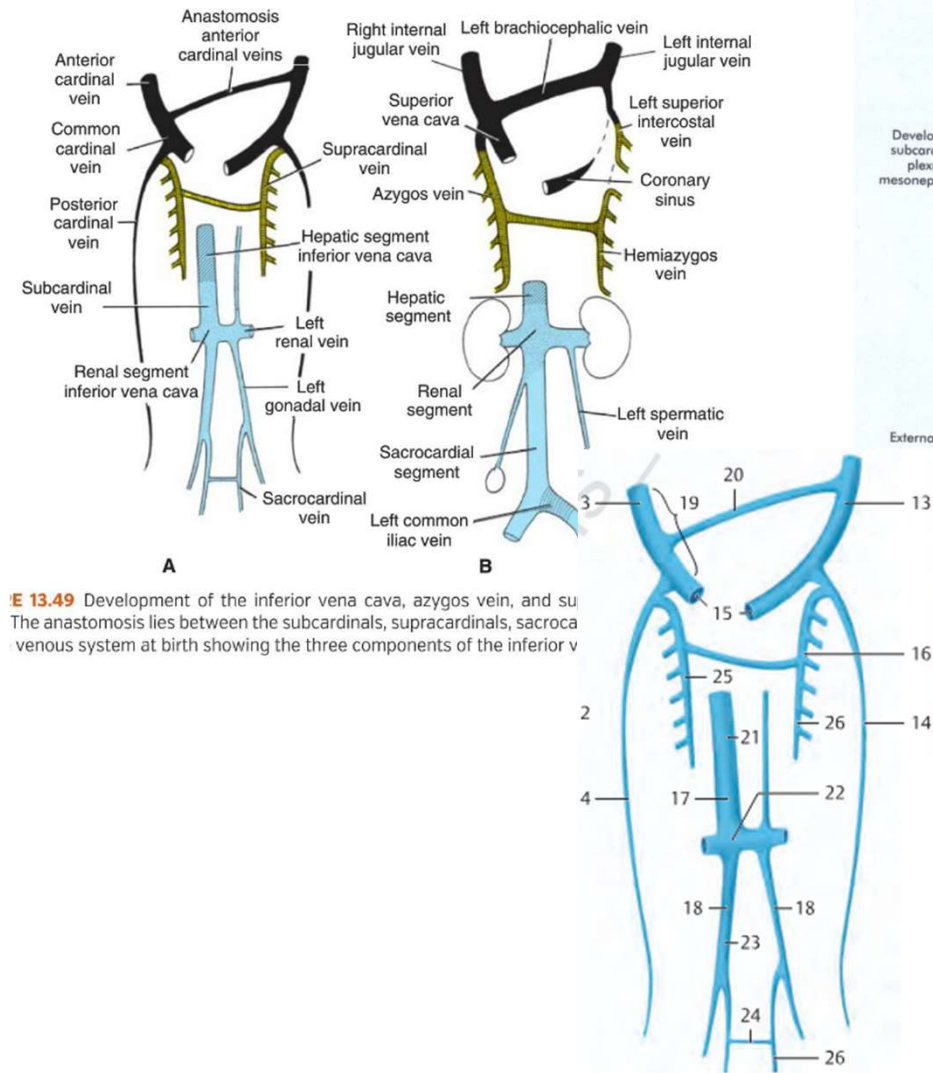




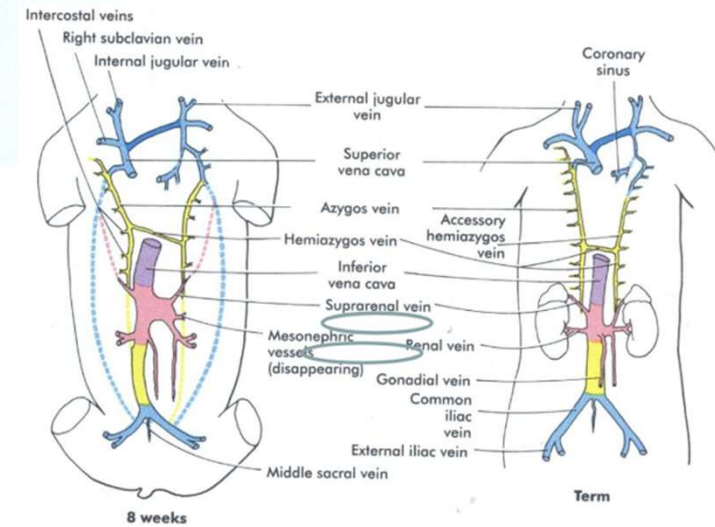
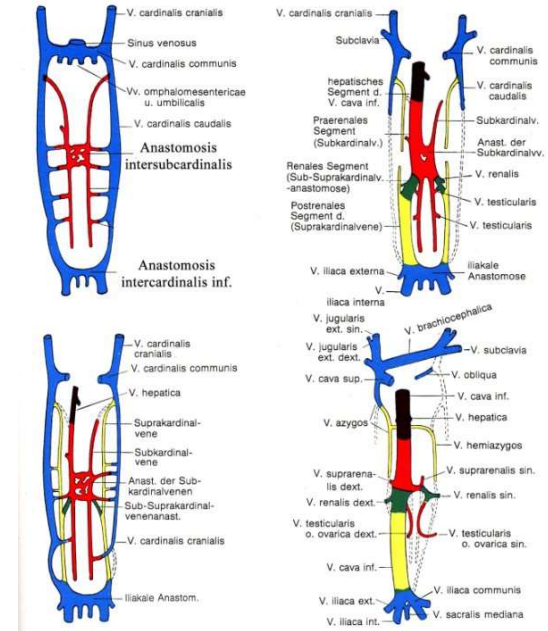
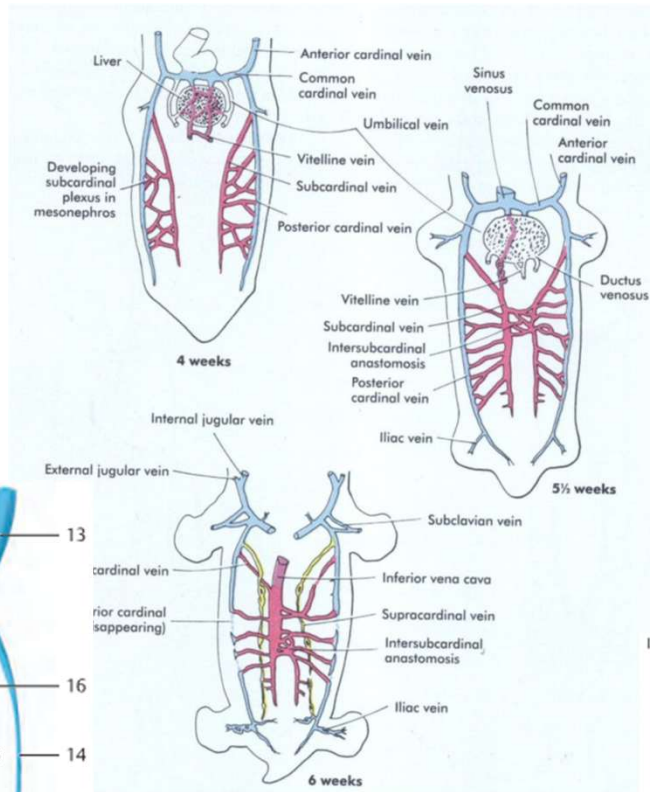
INFERIOR VENA CAVA



Controversies of caudal venous development



E 13.49 Development of the inferior vena cava, azygos vein, and subcardinal veins. The anastomosis lies between the subcardinals, supracardinals, sacrocardinal, and sacrocaudal venous system at birth showing the three components of the inferior v



2015 Development of the human infrahepatic inferior caval and azygos venous systems Jill P. J. M. Hikspoors, 1 Jelly H. M. Soffers, 1 Hayelom K. Mekonen,1 Pieter Cornillie, 2 S. Eleonore Köhler 1 and Wouter H. Lamers 1,3 1Department of Anatomy & Embryology, Maastricht University, Maastricht, The Netherlands 2Department of Morphology, Faculty of Veterinary Medicine, Ghent University, Merelbeke, Belgium 3Tygat Institute for Liver and Intestinal Research, Academic Medical Centre, Amsterdam, The Netherlands

Historical findings:
Older methodology,
Possible misinterpretations

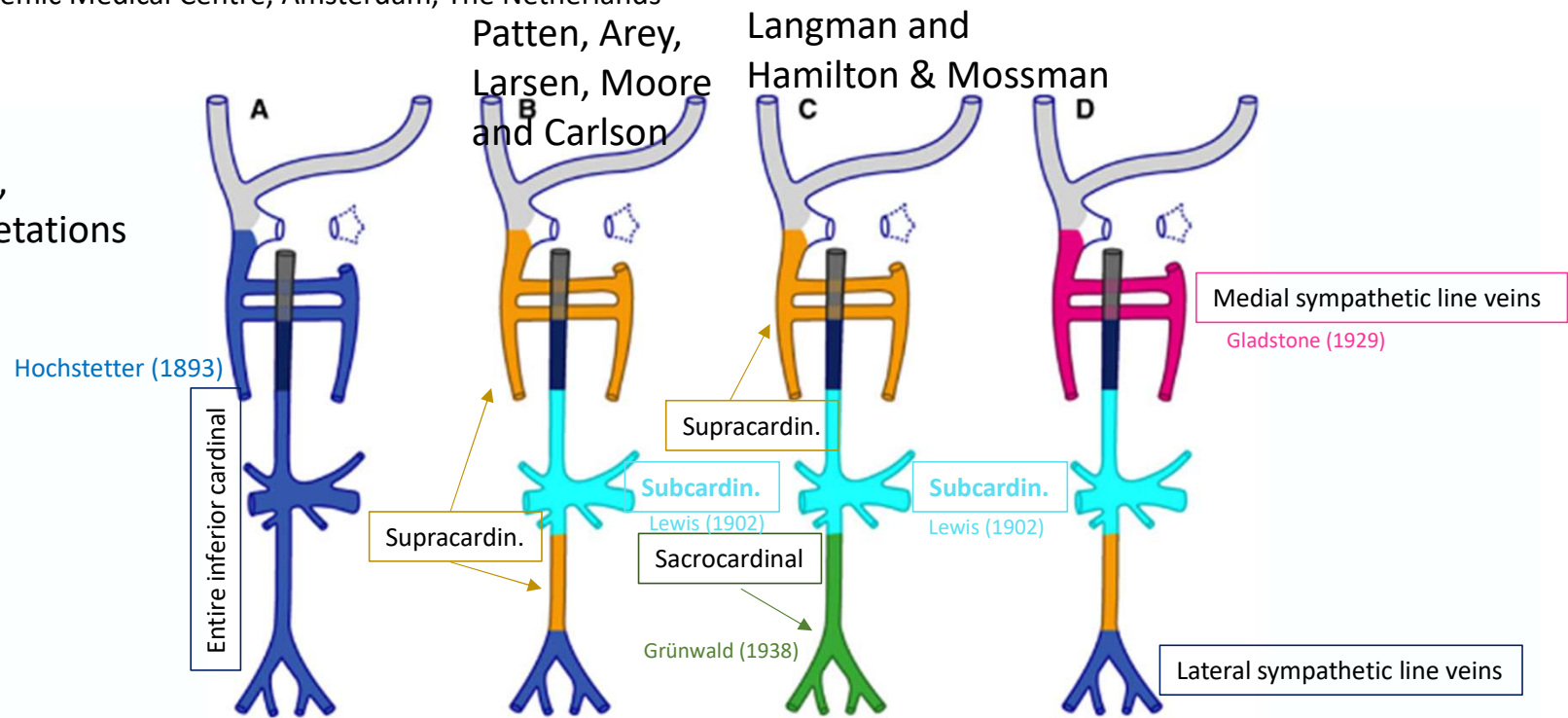


Fig. 1 Different theories explaining inferior caval and azygos vein development. The scheme represents the adult configuration. The colour codes indicate the developmental origin of the veins according to the different theories on caudal vein development: light grey – superior caval and jugular veins; blue – derived from caudal cardinal veins; cyan – subcardinal veins; orange – supracardinal or lateral sympathetic line veins; pink – medial sympathetic line veins; green – sacrocardinal veins. The intrahepatic part of the inferior caval vein is shown in very dark blue and the hepatocardiac vein in grey. In (A), the entire caudal venous system originates from the caudal cardinal veins. In (B) the infrarenal part of the inferior caval and azygos venous systems originate from supracardinal veins, while the renal and associated veins are derived from the subcardinal system. (C) differs from (B) in that the infrarenal caval and iliac veins derive from the sacrocardinal veins. (D) is identical to (B) except that the azygos veins originate from the medial sympathetic line veins. In this last model, the infrarenal part originates from the supracardinal veins, which are now called lateral sympathetic line veins.

"This description shows that the **infrarenal part of the inferior caval vein** originated from **the (right) caudal cardinal vein**, whereas the **renal part originated from subcardinal veins**. The **azygos venous system** developed from the **cranial part of the caudal cardinal veins**. Based on the gradual expansion of the caudal cardinal veins and the gradual changes in the topography of its parts, **we see no need to invoke supracardinal, sacrocardinal or medial and lateral sympathetic line veins to explain infra- hepatic inferior caval and azygos vein development** and, therefore, reject the hypothesis that the caudal venous system is made up from several independently developing components."

3D digitalized reconstruction
 of H-E stained human
 embrionic serial sections

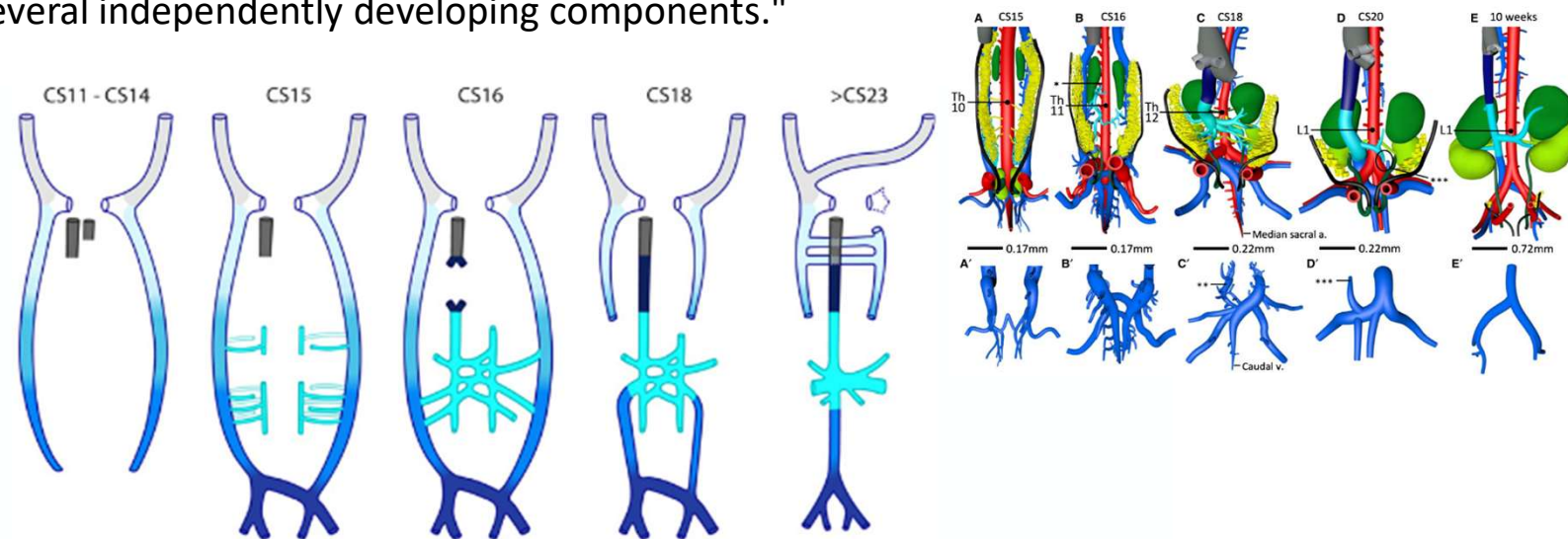


Fig. 2 Schematic view of the development of the inferior caval and azygos venous systems in human. Colour code: light grey: cranial cardinal veins; gradient from light to dark blue – caudal extension of the caudal cardinal veins with time; cyan – subcardinal veins, with vessels dorsal (open) and ventral (filled) to the mesonephroi indicated separately; very dark blue –intrahepatic; grey – hepatocardiac vein. The essence of this 'cardinal model' is that the caudal cardinal veins first extend caudally between CS11 and CS15, and then form a ventral plexus around the mesonephroi at CS15. Between CS16 and CS20, the caudal extension transforms into the iliac veins and their confluence into the inferior caval vein, while the ventral extension transforms into the subcardinal veins and then the renal and associated veins. Major changes in the caudal cardinal veins are also their interruption at the level of the future diaphragm at CS18 and the disappearance of the left caudal part of the caudal cardinal vein at CS20. Between CS18 and CS23, the azygos venous system develops from the cranial part of the caudal cardinal veins.

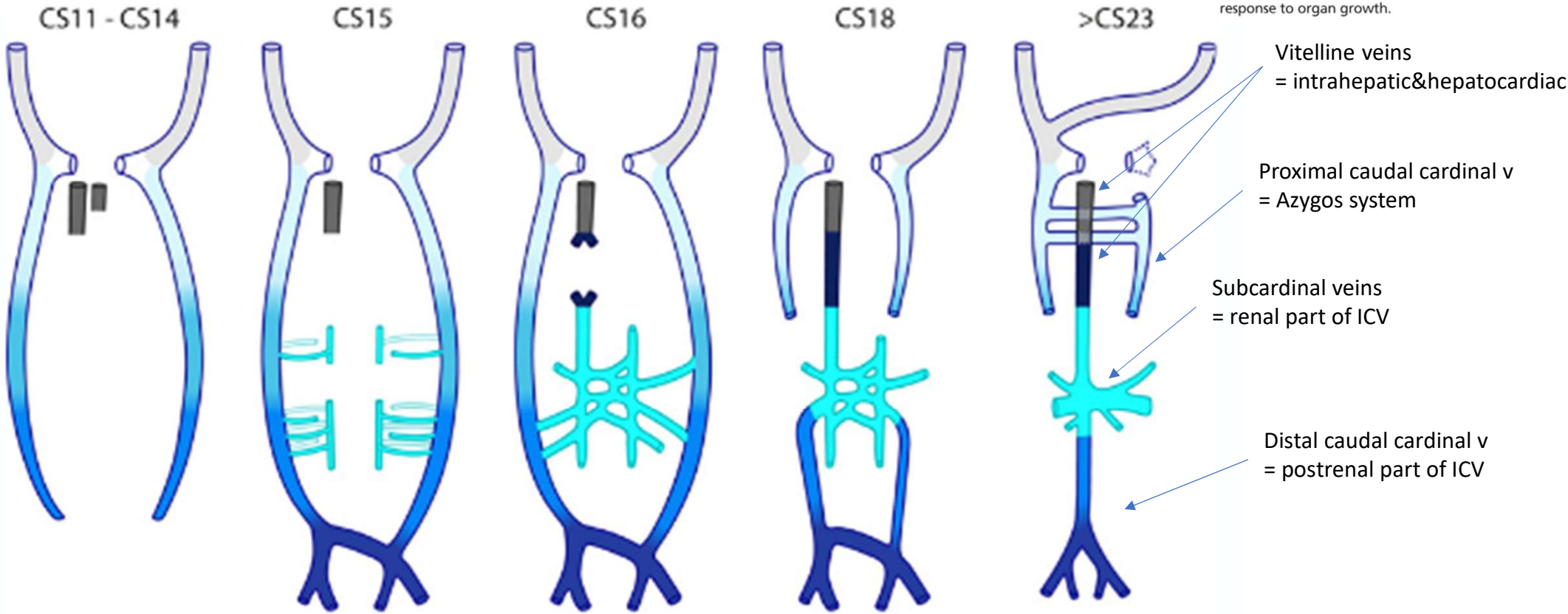
Development of the human infrahepatic inferior caval and azygos venous systems Jill P. J. M. Hikspoors, 1 Jelly H. M. Soffers, 1 Hayelom K. Mekonen,1 Pieter Cornillie, 2 S. Eleonore K€ohler 1 and Wouter H. Lamers 1,3 1Department of Anatomy & Embryology, Maastricht University, Maastricht, The Netherlands 2Department of Morphology, Faculty of Veterinary Medicine, Ghent University, Merelbeke, Belgium 3Tygat Institute for Liver and Intestinal Research, Academic Medical Centre, Amsterdam, The Netherlands

10.1111/JOA.12266

2015

Conclusion

We reinvestigated the development of the infrahepatic part of the inferior caval and azygos venous systems in human to distinguish concepts that account for developmental changes in the appearance of the caudal venous system. Our findings, which are summarized in Fig. 2, clearly support a module in which the infrahepatic part of the inferior caval vein develops from contributions of the caudal cardinal veins or from the subcardinal veins, i.e. a plexus sprouting from the caudal cardinal veins. The azygos venous system represents the cranial part of the caudal cardinal veins. The caudal cardinal and subcardinal veins remodel continuously due to adaptive changes in their position in response to organ growth.



*Far fewer names

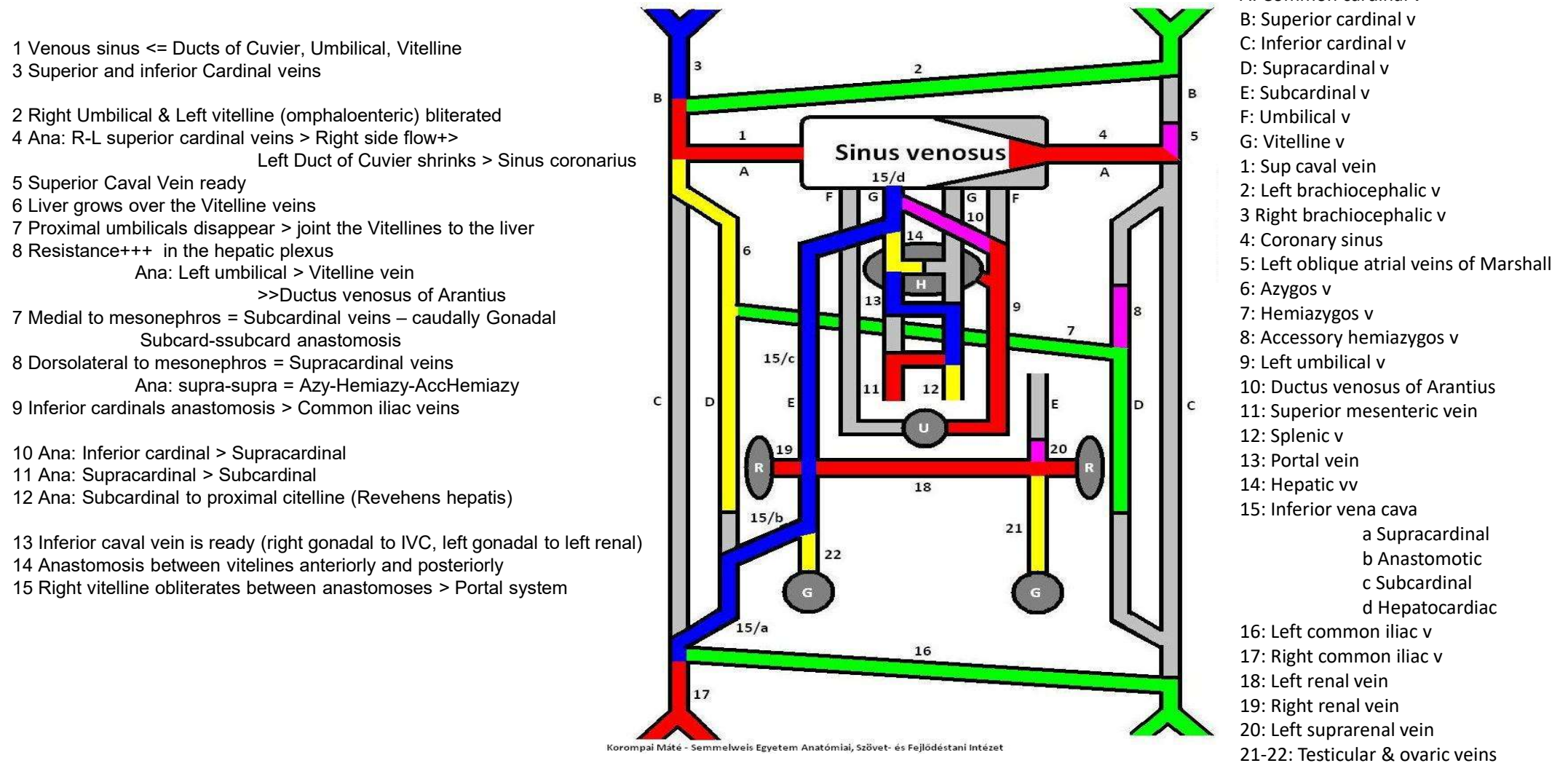


Yes, very sad

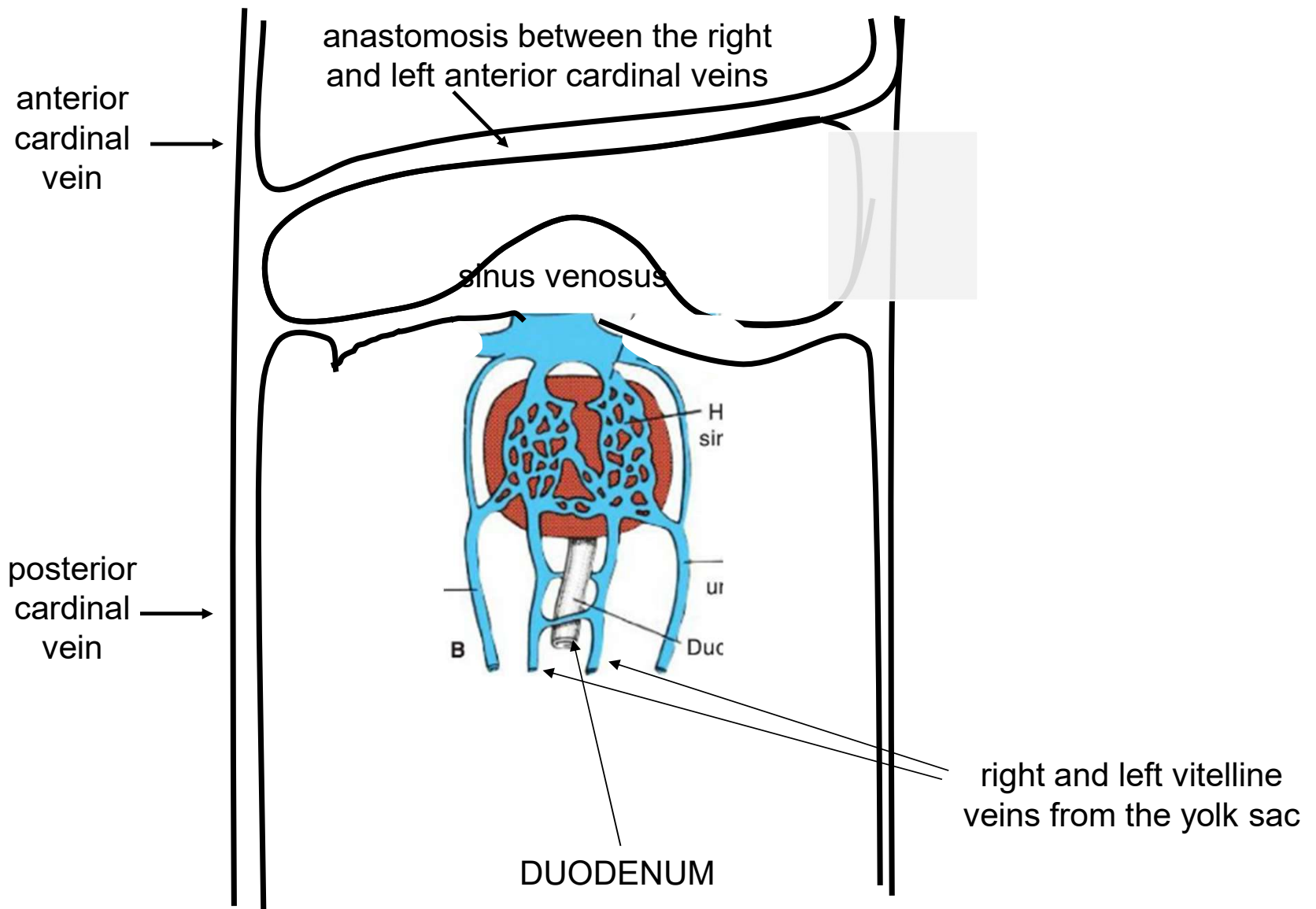
Anyway...

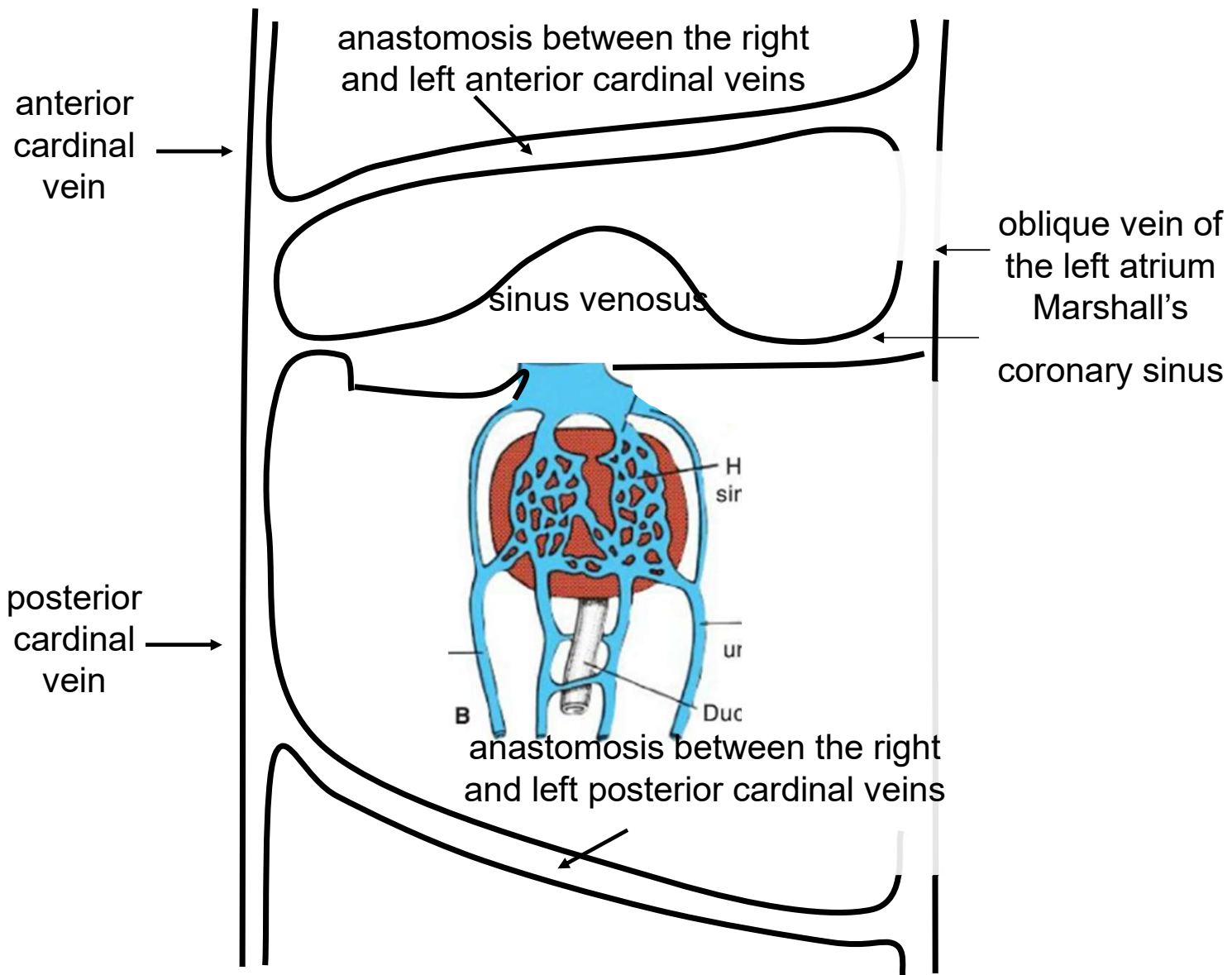
We will **continue to refer to the segments by these previously used names**, as they will help us better understand the morphology of the process. (The questions on the test are based on those names – and in respect of tradition.)

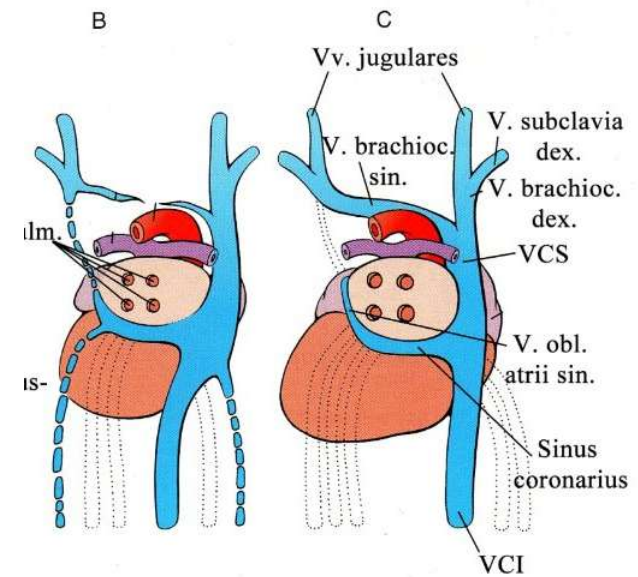
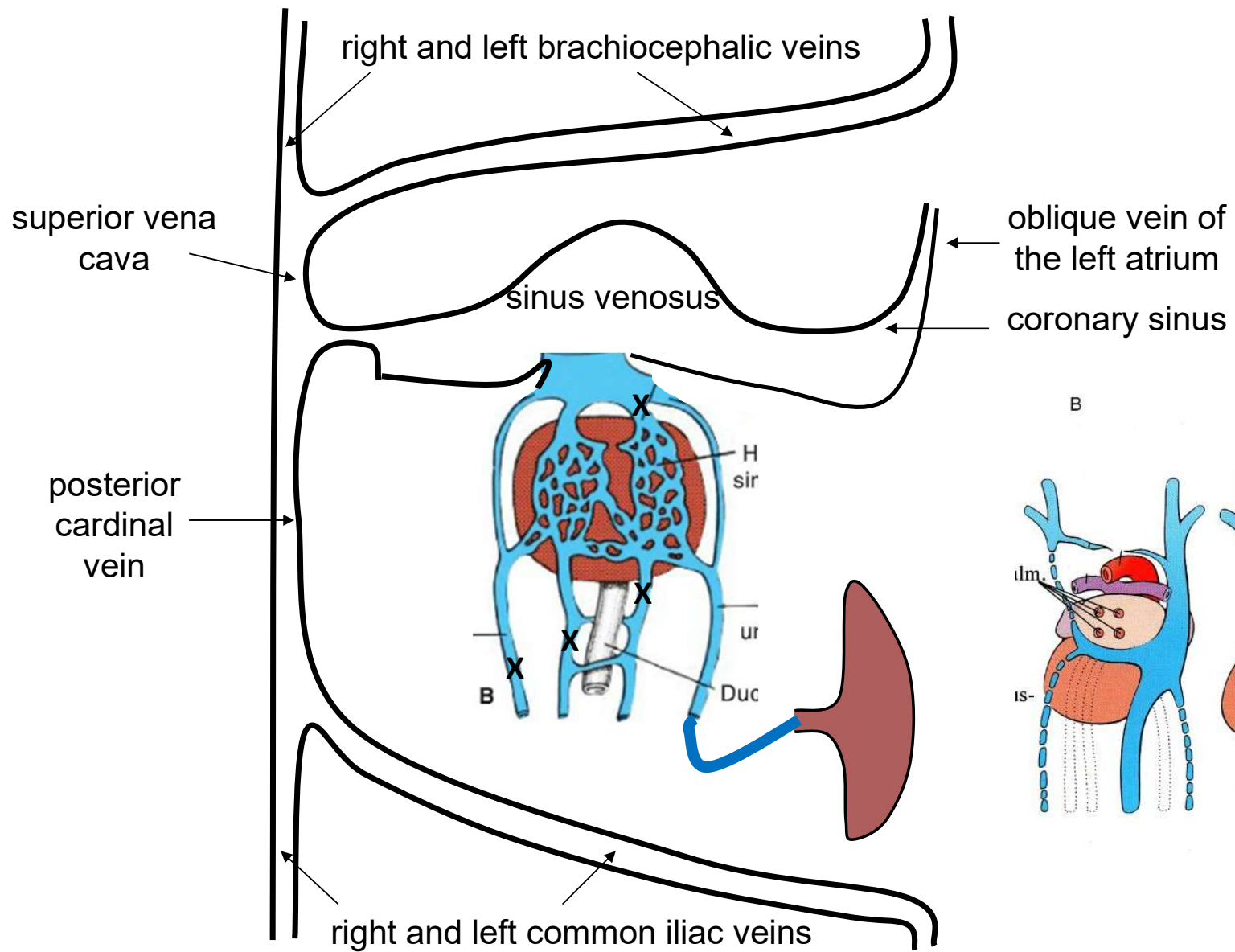
Major events during venous development

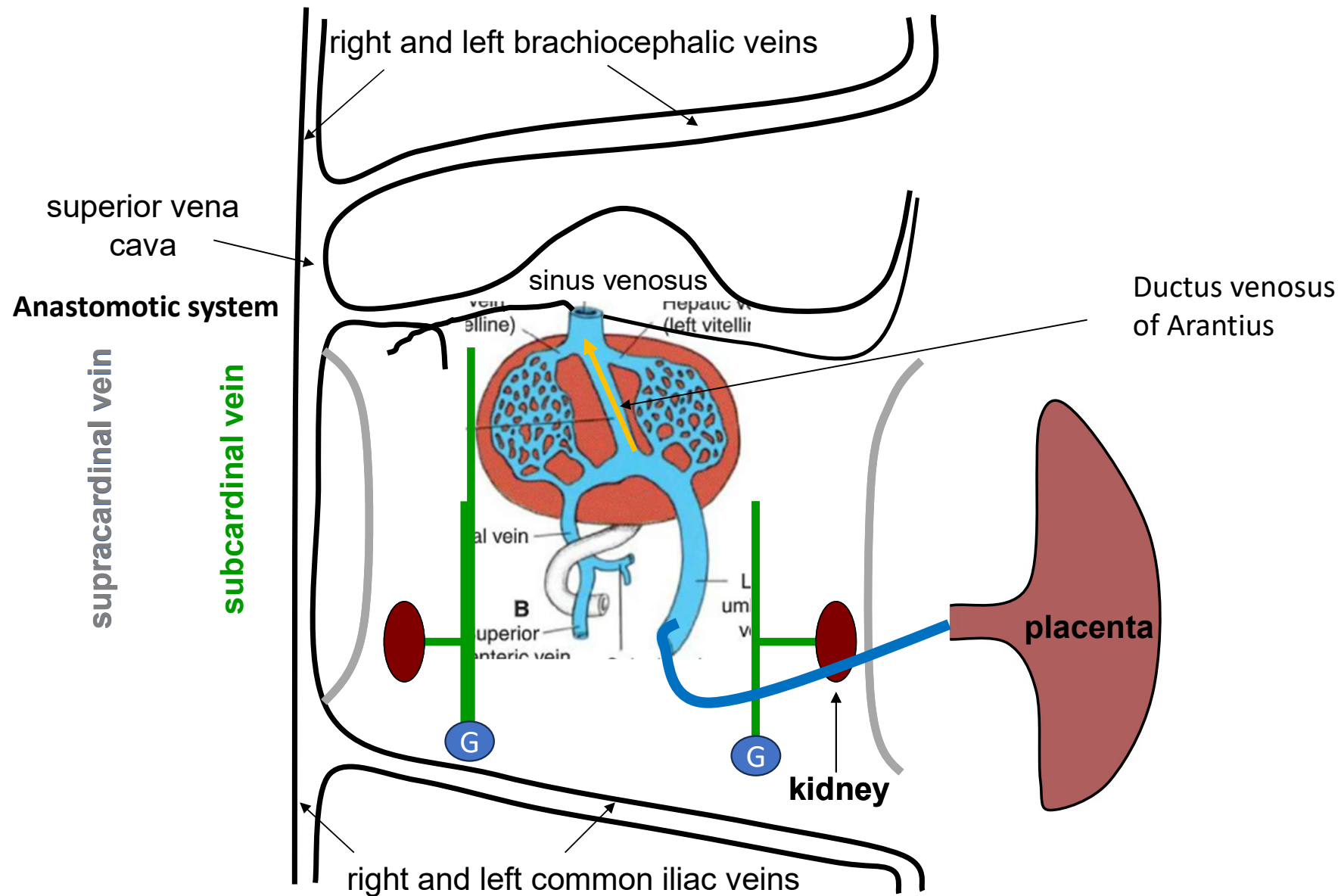


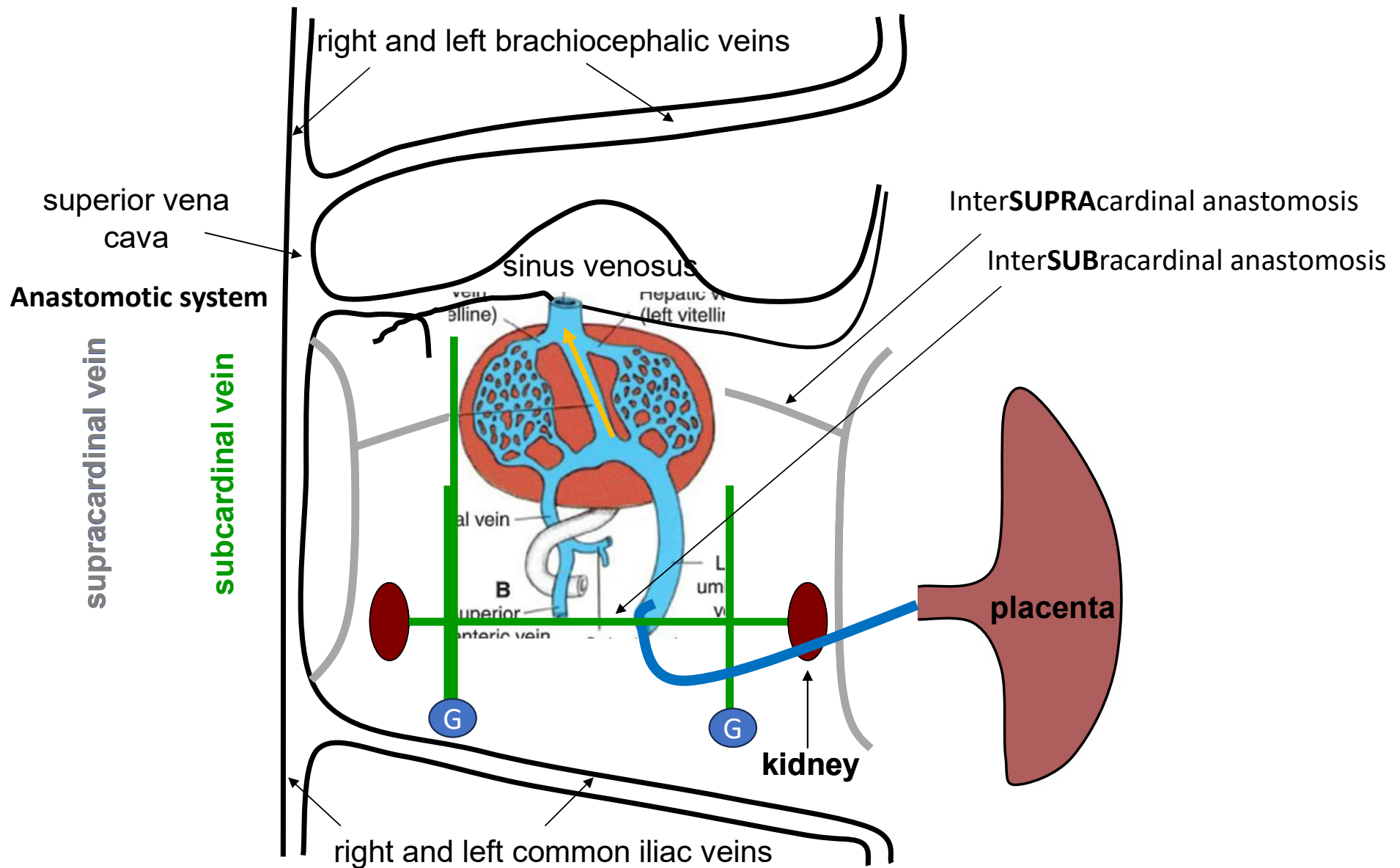
Korompai Máté's fig.

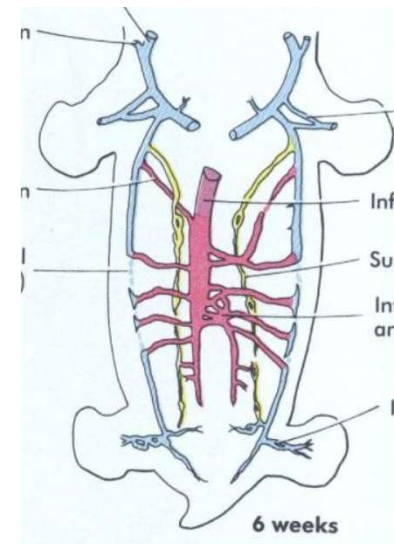
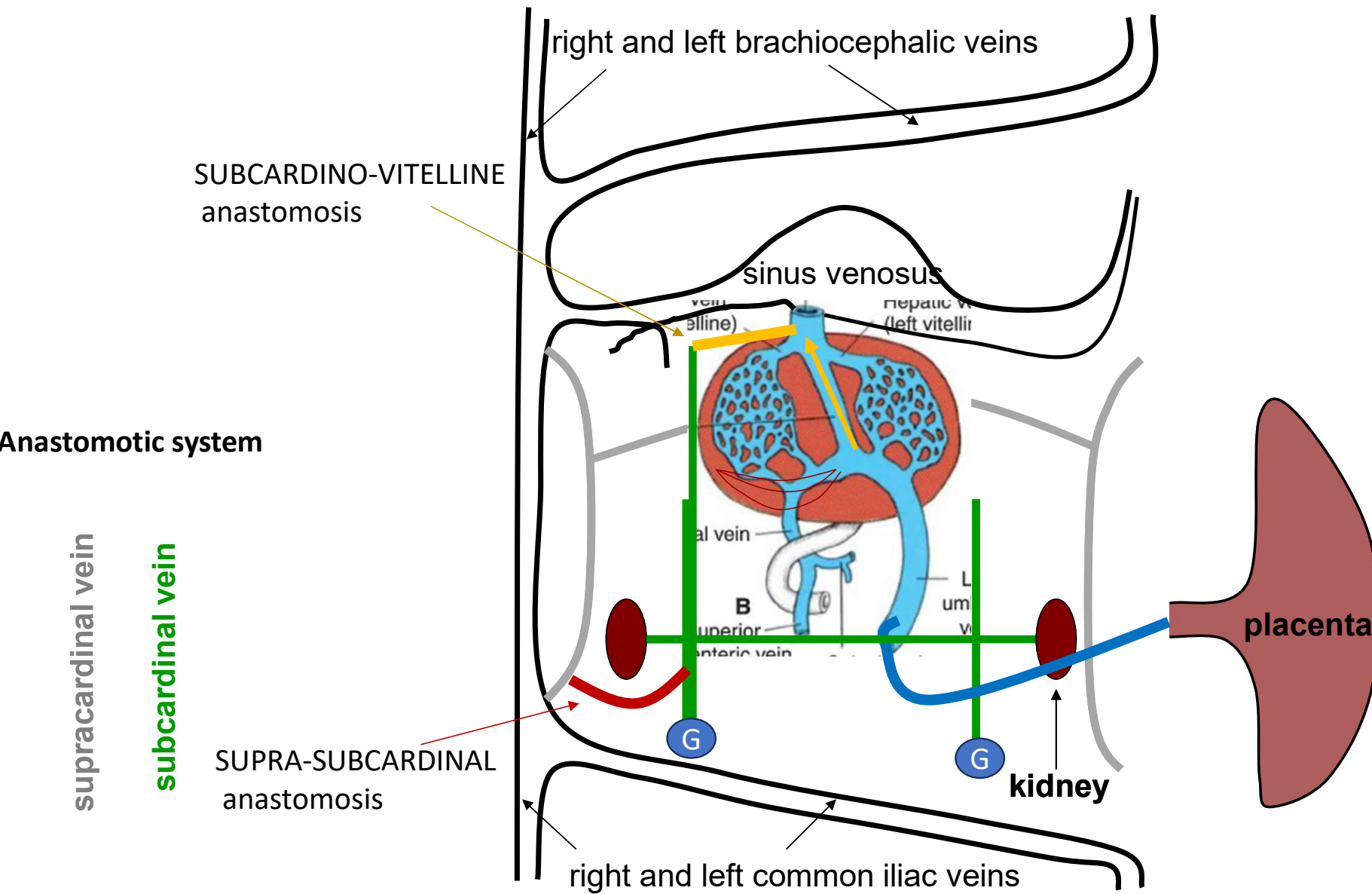












**Venous
Anastomotic
system**

Formation of the inferior vena cava

superior vena cava
Anastomotic system

Formation of the inferior vena cava

Hepatocardiac part

right vitelline

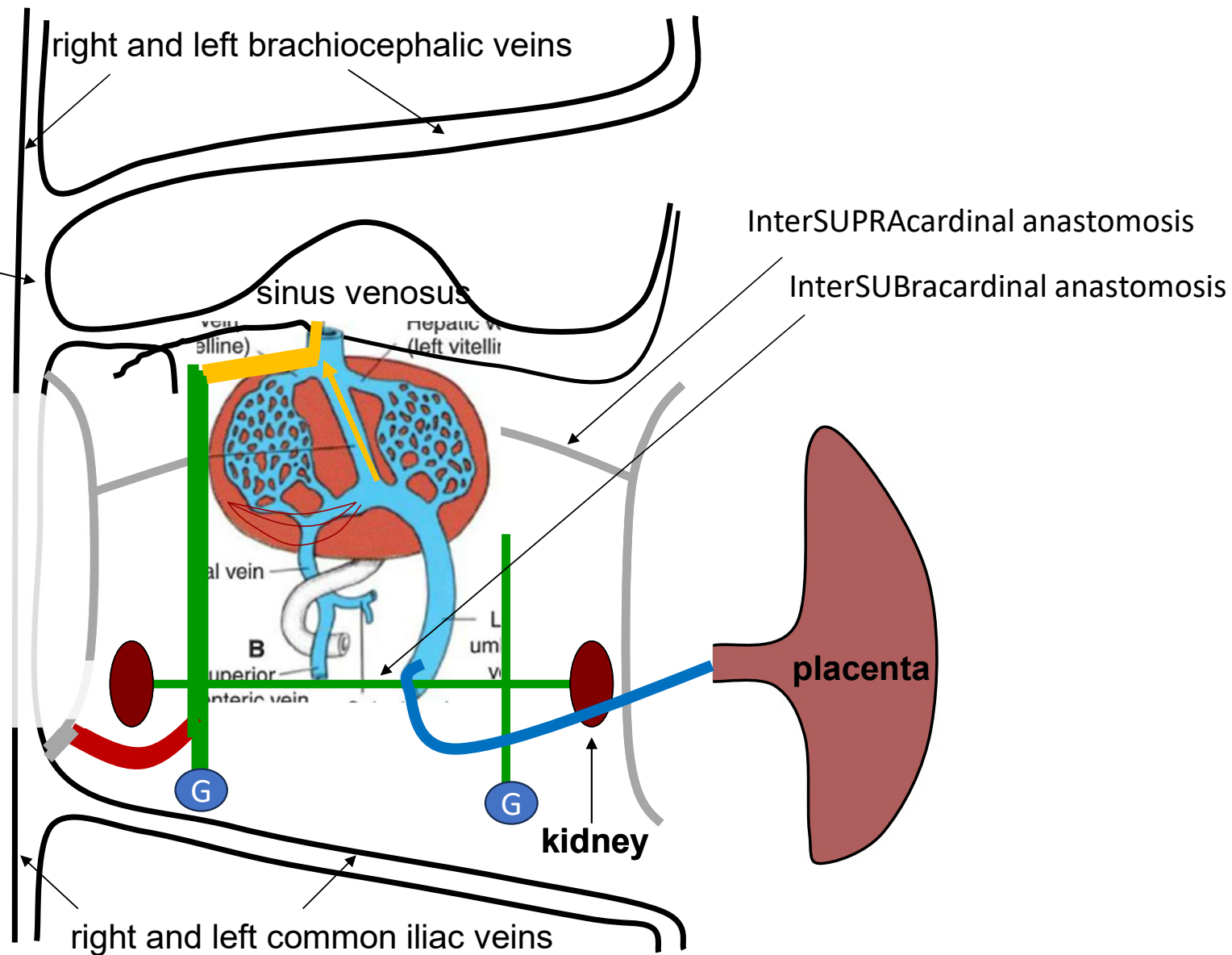
Renal part

R- L subcardinal + anastomoses

Postrenal segment

right supracardinal + sub-supracard anast.

supracardinal vein
subcardinal vein



Formation of the inferior vena cava

Formation of the inferior vena cava

Hepatocardiac part

right vitelline

Renal part

R- L subcardinal + anastomoses

Postrenal segment

right supracardinal + sub-supracard anast.

azygos vein

Inf. cardinal + right
supracardinal

sinus venosus

Lig. venosum

Round ligament

AZYGOS SYSTEM

&

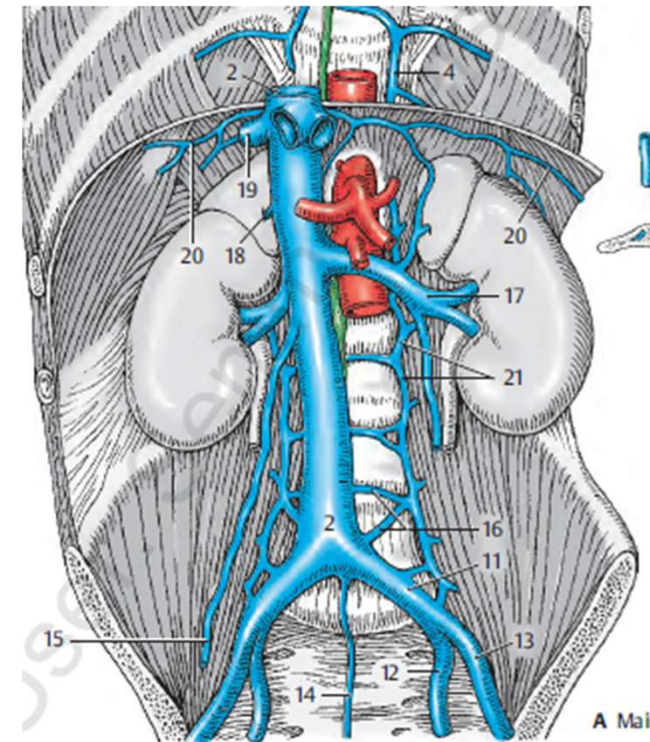
RUDIMENTARY UMBILICAL RELATED LIGAMENTS

hemiazygos vein

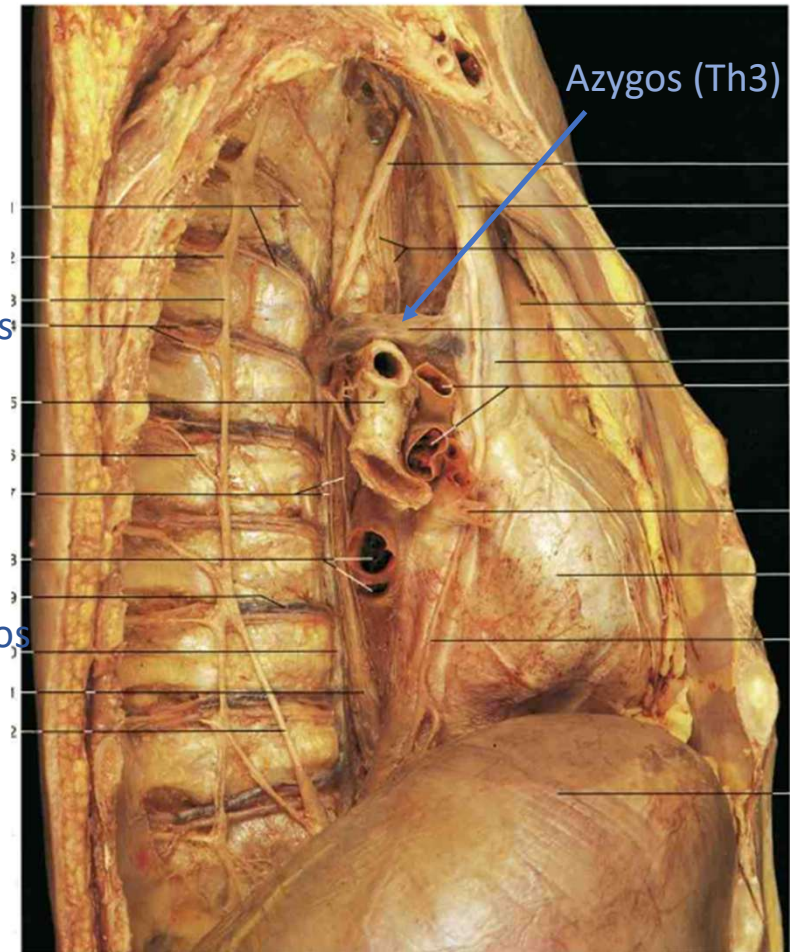
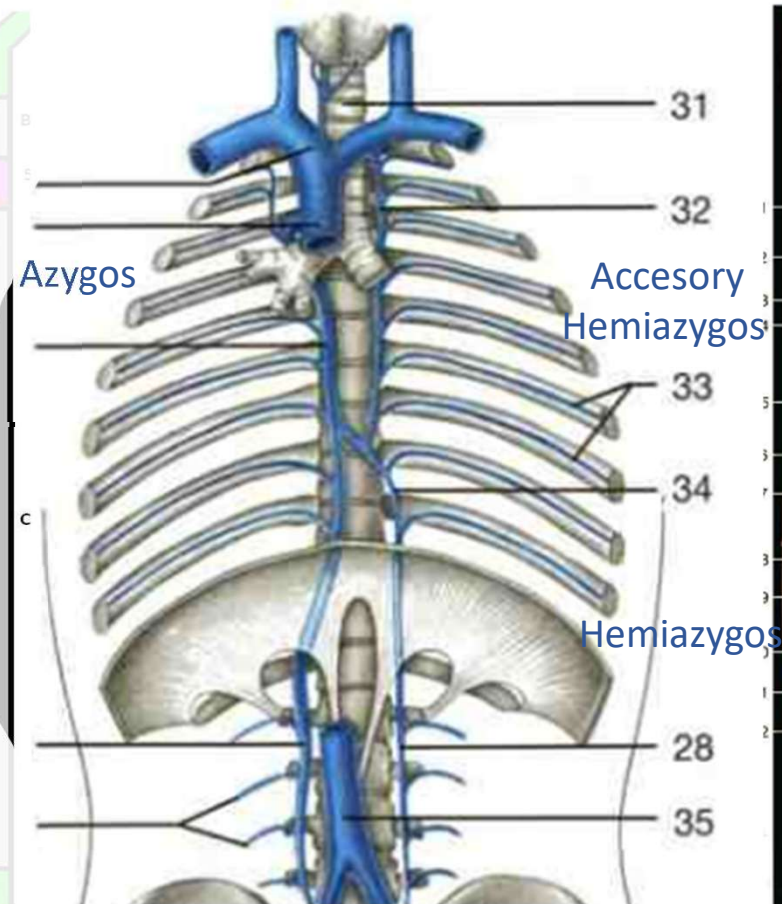
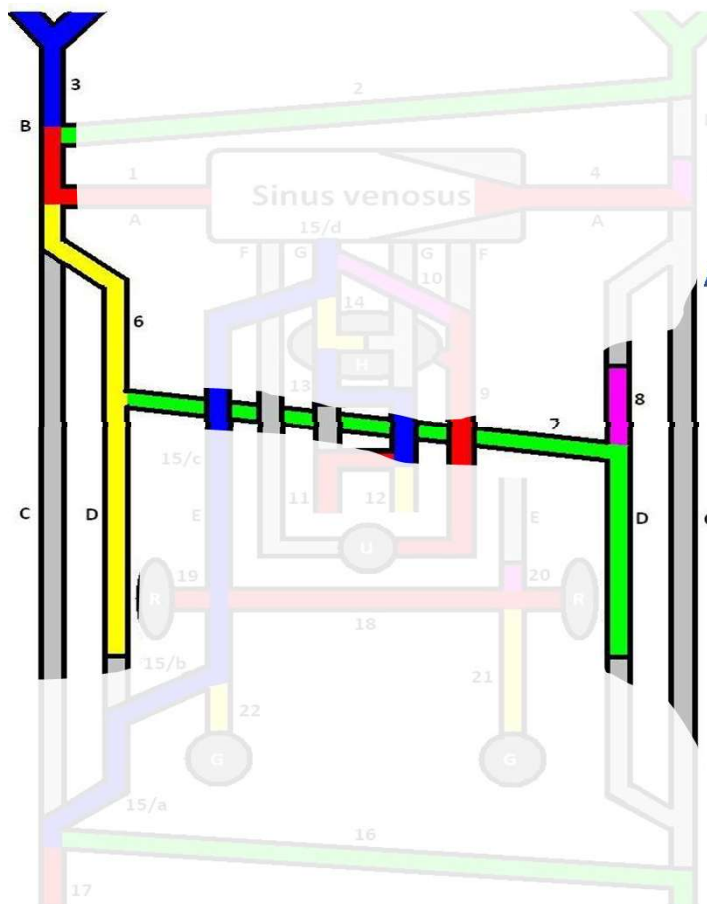
(anastomosis, and lower left supracardinal)

accessory hemiazygos vein

(left supracardinal supranastomotic part)



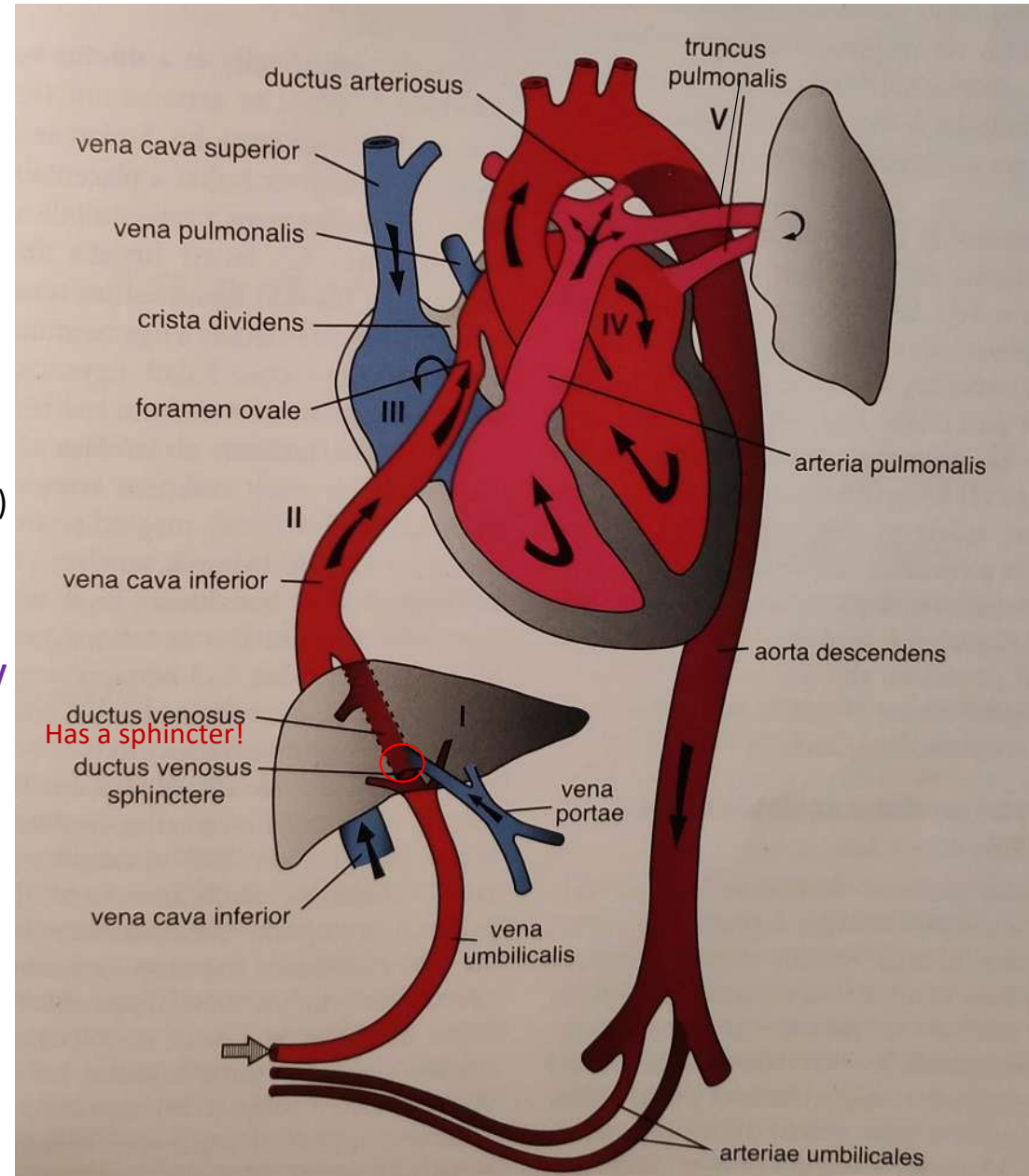
Supracardinal veins to Azygos system



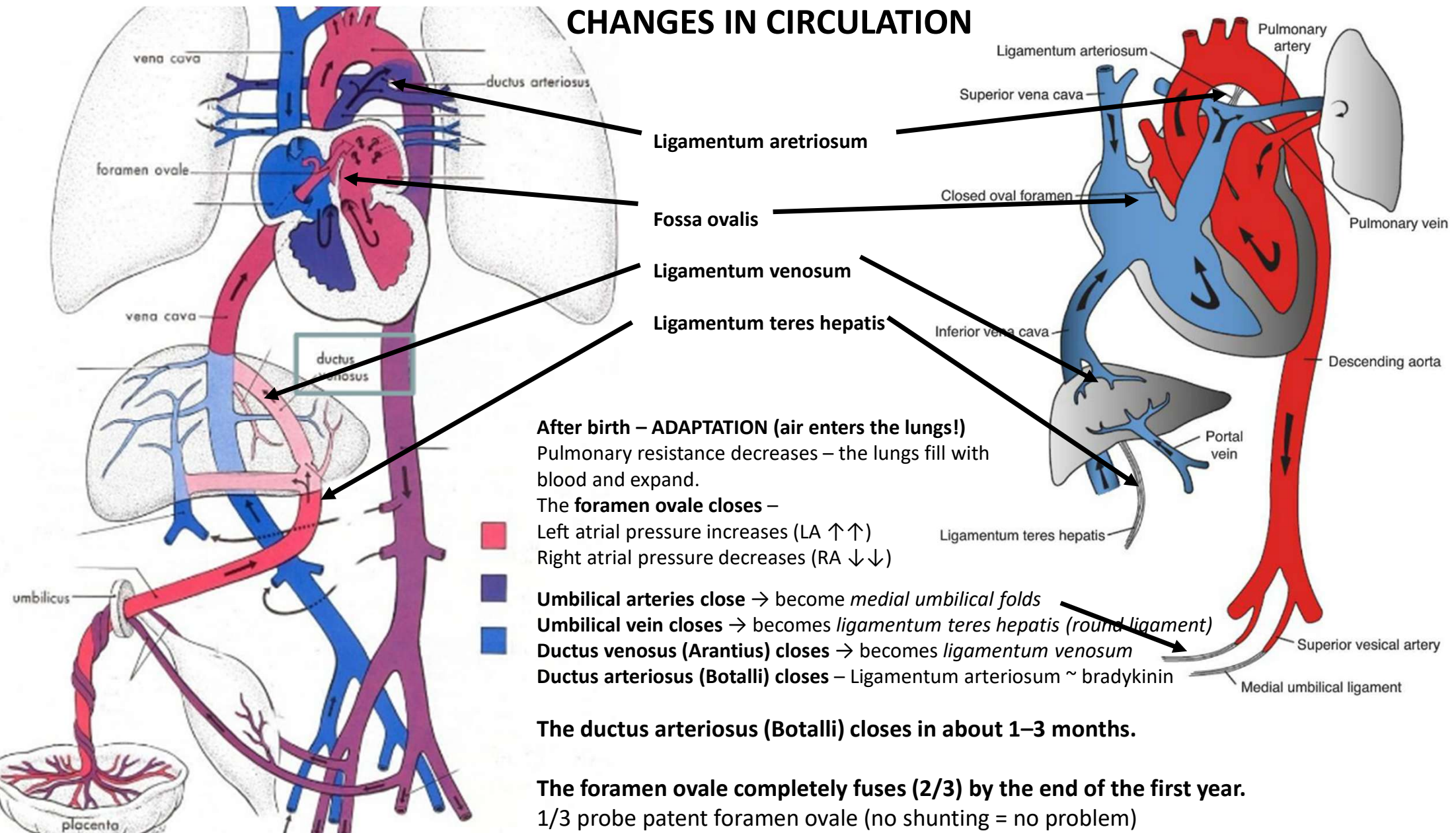
Fetal circulation

Placenta → oxygenated blood

- Umbilical vein
- Ductus venosus (Arantius)
- Inferior vena cava (IVC)
- Right atrium (+superior vena cava mostly to RV > Pulm.artery > Botallo)
- Foramen ovale (Still well oxygenated)
- Left atrium → Aorta
- **First to the heart and brain**
- **Ductus arteriosus of Botallo carries** deoxygenated blood > **Lower body**
- Umbilical arteries
- Placental villi
- Oxygenation
- Umbilical vein



CHANGES IN CIRCULATION



- In the early stage of the embryonic development a pair of vitelline veins carry blood from the yolk sac, but later only the right one persists.
- With the appearance and progressive development of the placenta the role of the vitelline veins is replaced by a pair of umbilical veins, but only the left one remains between the placenta and the primordial heart.
- The main venous system draining the head and the caudal part of the embryo is represented by the anterior and posterior cardinal veins, respectively.
- Supracardinal veins appear on the lateral side of the developing kidney. The supracardinal veins are connected with a horizontal anastomosis, resulting in a H-shaped structure.
 - On the medial sides of the primordial kidneys a new pair of veins develop: the subcardinal veins, which are also interconnected with a horizontal anastomosis.
 - The liver developing in the septum transversum divides the remaining right vitelline vein in three segments: prehepatic, intrahepatic and posthepatic parts can be distinguished.
 - Due to the growing liver the left umbilical vein is interrupted and loses its contact with the heart. The larger proportion of the blood arriving from the placenta enters the inferior vena cava on the surface of the liver via the ductus venosus, while the rest enters the hepatic sinuses.

- On the left side the cardinal veins degenerate. The only notable persisting vessel is the coronary sinus.
- The right anterior cardinal vein develops into the right brachiocephalic vein and the superior vena cava, while the anastomosis between the two anterior cardinal veins remains as the left brachiocephalic vein.
- Supra- and subcardinal veins with the prehepatic segment of the right vitelline vein and the interconnecting anastomoses play an important role in formation of the inferior vena cava.
- Prehepatic segment of the right vitelline vein develops into the portal vein.
- The major part of the supracardinal veins persists as the vena azygos/hemiazygos system.
- Subcardinal veins are incorporated into the inferior vena cava or develop into the renal and gonadal veins.

Sources:

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